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Harvesting energy for a smart ring via piezoelectric charging

Abstract

A smart ring includes a housing configured to be worn by a user. The smart ring can further include a power source disposed within the housing. The smart ring also can include a charging circuit disposed within the housing. The charging circuit can include a harvesting element configured to charge the power source. The harvesting element can be configured to convert mechanical energy to electrical energy. The smart ring also can include a device at least partially disposed within the housing. The device can be configured to draw energy from the power source. Other embodiments are disclosed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
513344	12/1893	Smith	N/A	N/A
2737192	12/1955	Bieler	N/A	N/A
3792192	12/1973	Plate	N/A	N/A
4012629	12/1976	Simms	N/A	N/A
4382112	12/1982	Betts et al.	N/A	N/A
4684687	12/1986	Breach et al.	N/A	N/A
4830014	12/1988	Goodman et al.	N/A	N/A
4880304	12/1988	Jaeb et al.	N/A	N/A
5135220	12/1991	Baldoni	N/A	N/A
6097480	12/1999	Kaplan	N/A	N/A
6154658	12/1999	Caci	N/A	N/A
6201698	12/2000	Hunter	N/A	N/A
6560993	12/2002	Bosque et al.	N/A	N/A
6608562	12/2002	Kimura et al.	N/A	N/A
6699199	12/2003	Asada et al.	N/A	N/A
6745061	12/2003	Hicks et al.	N/A	N/A
6792044	12/2003	Peng et al.	N/A	N/A
6800693	12/2003	Nishihara et al.	N/A	N/A
6803391	12/2003	Paglia et al.	N/A	N/A
6805140	12/2003	Velez, Jr. et al.	N/A	N/A
6894628	12/2004	Marpe et al.	N/A	N/A
6909753	12/2004	Meehan et al.	N/A	N/A
6959116	12/2004	Sezer et al.	N/A	N/A
7013674	12/2005	Kretchmer	N/A	N/A
7136532	12/2005	VanDer	N/A	N/A
7190986	12/2006	Hannula et al.	N/A	N/A
7227894	12/2006	Lin et al.	N/A	N/A
7275172	12/2006	Janzen	N/A	N/A
7286710	12/2006	Marpe et al.	N/A	N/A
7468036	12/2007	Rulkov et al.	N/A	N/A
7500746	12/2008	Howell et al.	N/A	N/A
7519229	12/2008	Wallace et al.	N/A	N/A
7689437	12/2009	Teller et al.	N/A	N/A
7872444	12/2010	Hamilton et al.	N/A	N/A
8031172	12/2010	Kruse et al.	N/A	N/A
8075484	12/2010	Moore-Ede	N/A	N/A
8345752	12/2012	Lee et al.	N/A	N/A
8446275	12/2012	Utter, II	N/A	N/A
8554297	12/2012	Moon et al.	N/A	N/A
8570273	12/2012	Smith	N/A	N/A
8700111	12/2013	Leboeuf et al.	N/A	N/A
8954135	12/2014	Yuen et al.	N/A	N/A
9218058	12/2014	Bress et al.	N/A	N/A
9248839	12/2015	Tan	N/A	N/A
9248938	12/2015	Hopps	N/A	N/A
9265310	12/2015	Lam	N/A	N/A
9362775	12/2015	Jacobs	N/A	N/A
9420260	12/2015	McGregor et al.	N/A	N/A
9440657	12/2015	Fields et al.	N/A	N/A
9477146	12/2015	Xu et al.	N/A	N/A
9509170	12/2015	Wu	N/A	N/A

9628707	12/2016	Blum et al.	N/A	N/A
9660488	12/2016	Breedvelt-Schouten et al.	N/A	N/A
9696690	12/2016	Nguyen et al.	N/A	N/A
9711060	12/2016	Lusted et al.	N/A	N/A
9711993	12/2016	Kim	N/A	N/A
9717949	12/2016	Tran et al.	N/A	N/A
9756301	12/2016	Li et al.	N/A	N/A
9841331	12/2016	Wood et al.	N/A	N/A
9847020	12/2016	Davis	N/A	N/A
9861314	12/2017	Haverinen et al.	N/A	N/A
9908530	12/2017	Fields et al.	N/A	N/A
9931976	12/2017	Terwilliger et al.	N/A	N/A
9955286	12/2017	Segal	N/A	N/A
9956963	12/2017	Vijaya Kumar et al.	N/A	N/A
9965761	12/2017	Elangovan et al.	N/A	N/A
10007355	12/2017	Schorsch et al.	N/A	N/A
10085695	12/2017	Ouwerkerk et al.	N/A	N/A
10099608	12/2017	Cuddihy et al.	N/A	N/A
10102510	12/2017	Yau et al.	N/A	N/A
10137777	12/2017	Lu et al.	N/A	N/A
10139859	12/2017	Von Badinski et al.	N/A	N/A
10281953	12/2018	Von Badinski	N/A	N/A
10303867	12/2018	Schröder	N/A	N/A
10315557	12/2018	Terwilliger et al.	N/A	N/A
10317940	12/2018	Eim et al.	N/A	N/A
10345506	12/2018	Lyu	N/A	N/A
10359846	12/2018	Priyantha et al.	N/A	N/A
10366220	12/2018	Shapiro et al.	N/A	N/A
10377386	12/2018	Richmond	N/A	N/A
10396584	12/2018	Madau et al.	N/A	N/A
10409327	12/2018	Stotler	N/A	N/A
10444834	12/2018	Vescovi et al.	N/A	N/A
10463141	12/2018	Fitzgerald et al.	N/A	N/A
10528989	12/2019	Irey	N/A	N/A
10564628	12/2019	Hargovan et al.	N/A	N/A
10629175	12/2019	Yan et al.	N/A	N/A
10664842	12/2019	Bermudez et al.	N/A	N/A
10693872	12/2019	Larson et al.	N/A	N/A
10703204	12/2019	Hassan et al.	N/A	N/A
10709339	12/2019	Lusted	N/A	N/A
10745032	12/2019	Scheggi	N/A	N/A
10762183	12/2019	Charan et al.	N/A	N/A
10768666	12/2019	Von Badinski et al.	N/A	N/A
10838499	12/2019	Wang	N/A	N/A
10842429	12/2019	Kinnunen et al.	N/A	N/A
10849557	12/2019	Keating et al.	N/A	N/A
10893833	12/2020	Haverinen et al.	N/A	N/A
10944745	12/2020	Kursun et al.	N/A	N/A
11227060	12/2021	John et al.	N/A	N/A
11265635	12/2021	Shankar	N/A	N/A
11312299	12/2021	Assam	N/A	N/A
11479258	12/2021	Sanchez	N/A	N/A
11599147	12/2022	Von Badinski et al.	N/A	N/A
11637511	12/2022	Sanchez	320/107	H02N 2/18
11868178	12/2023	Von Badinski et al.	N/A	N/A
11868179	12/2023	Von Badinski et al.	N/A	N/A
11894704	12/2023	Sanchez	N/A	N/A
11909238	12/2023	Sanchez	N/A	N/A
11923791	12/2023	Sanchez	N/A	H02J 7/007
11984742	12/2023	Sanchez	N/A	N/A
12191692	12/2024	Sanchez	N/A	N/A

12211467	12/2024	Sanchez	N/A	N/A
12237700	12/2024	Sanchez et al.	N/A	N/A
2002/0042464	12/2001	Barclay et al.	N/A	N/A
2002/0091568	12/2001	Kraft et al.	N/A	N/A
2002/0121831	12/2001	Egawa et al.	N/A	N/A
2003/0077064	12/2002	Katayama	N/A	N/A
2003/0142065	12/2002	Pahlavan	N/A	N/A
2004/0102551	12/2003	Sato et al.	N/A	N/A
2004/0118592	12/2003	Pehlert	N/A	N/A
2004/0145256	12/2003	Miekka et al.	N/A	N/A
2004/0160635	12/2003	Ikeda et al.	N/A	N/A
2004/0200235	12/2003	Kretchmer	N/A	N/A
2005/0012648	12/2004	Marpe et al.	N/A	N/A
2005/0030205	12/2004	Konoshima et al.	N/A	N/A
2005/0054941	12/2004	Ting et al.	N/A	N/A
2005/0062454	12/2004	Raghunath et al.	N/A	N/A
2005/0133248	12/2004	Easter	N/A	N/A
2005/0162523	12/2004	Darrell et al.	N/A	N/A
2005/0185060	12/2004	Neven, Sr.	N/A	N/A
2005/0185843	12/2004	Kudoh	N/A	N/A
2005/0185844	12/2004	Ono et al.	N/A	N/A
2005/0230596	12/2004	Howell et al.	N/A	N/A
2006/0002607	12/2005	Boncyk et al.	N/A	N/A
2006/0069681	12/2005	Lauper	N/A	N/A
2006/0080286	12/2005	Svendsen	N/A	N/A
2006/0085477	12/2005	Phillips et al.	N/A	N/A
2006/0089792	12/2005	Manber et al.	N/A	N/A
2006/0211924	12/2005	Dalke et al.	N/A	N/A
2006/0250043	12/2005	Chung	N/A	N/A
2006/0271593	12/2005	De Mes et al.	N/A	N/A
2007/0149222	12/2006	Hodko et al.	N/A	N/A
2007/0159522	12/2006	Neven	N/A	N/A
2007/0188626	12/2006	Squilla et al.	N/A	N/A
2007/0200713	12/2006	Weber et al.	N/A	N/A
2007/0223826	12/2006	Ridge et al.	N/A	N/A
2008/0068559	12/2007	Howell et al.	N/A	N/A
2008/0136587	12/2007	Orr	N/A	N/A
2008/0174676	12/2007	Squilla et al.	N/A	N/A
2008/0218684	12/2007	Howell et al.	N/A	N/A
2008/0275309	12/2007	Stivoric et al.	N/A	N/A
2009/0056703	12/2008	Mills et al.	N/A	N/A
2010/0219989	12/2009	Asami et al.	N/A	N/A
2011/0007035	12/2010	Shai	N/A	N/A
2011/0080339	12/2010	Sun et al.	N/A	N/A
2011/0224875	12/2010	Cuddihy et al.	N/A	N/A
2012/0016245	12/2011	Niwa et al.	N/A	N/A
2012/0130203	12/2011	Stergiou et al.	N/A	N/A
2012/0184367	12/2011	Parrott et al.	N/A	N/A
2012/0218184	12/2011	Wissmar	N/A	N/A
2012/0293107	12/2011	Ajagbe	N/A	N/A
2012/0317024	12/2011	Rahman et al.	N/A	N/A
2013/0106603	12/2012	Weast et al.	N/A	N/A
2013/0211291	12/2012	Tran	N/A	N/A
2013/0335213	12/2012	Sherony et al.	N/A	N/A
2014/0107493	12/2013	Yuen et al.	N/A	N/A
2014/0118704	12/2013	Duelli et al.	N/A	N/A
2014/0120983	12/2013	Lam	N/A	N/A
2014/0187160	12/2013	Prencipe	N/A	N/A
2014/0218529	12/2013	Mahmoud et al.	N/A	N/A
2014/0238153	12/2013	Wood et al.	N/A	N/A
2014/0240132	12/2013	Bychkov	N/A	N/A

2014/0244009	12/2013	Mestas	N/A	N/A
2014/0274203	12/2013	Ganong, III	N/A	N/A
2014/0309849	12/2013	Ricci	N/A	N/A
2014/0361934	12/2013	Ely et al.	N/A	N/A
2014/0361945	12/2013	Misra et al.	N/A	N/A
2015/0003693	12/2014	Baca et al.	N/A	N/A
2015/0019266	12/2014	Stempora	N/A	N/A
2015/0028996	12/2014	Agrafioti	N/A	N/A
2015/0046996	12/2014	Slaby et al.	N/A	N/A
2015/0062086	12/2014	Nattukallingal	N/A	N/A
2015/0065090	12/2014	Yeh	N/A	N/A
2015/0098309	12/2014	Adams	N/A	N/A
2015/0124096	12/2014	Koravadi	N/A	N/A
2015/0126824	12/2014	Leboeuf et al.	N/A	N/A
2015/0158499	12/2014	Koravadi	N/A	N/A
2015/0186092	12/2014	Francis et al.	N/A	N/A
2015/0220109	12/2014	VonBadinski et al.	N/A	N/A
2015/0277559	12/2014	Vescovi et al.	N/A	N/A
2015/0338926	12/2014	Park et al.	N/A	N/A
2015/0352953	12/2014	Koravadi	N/A	N/A
2016/0028267	12/2015	Lee et al.	N/A	N/A
2016/0098530	12/2015	Dill et al.	N/A	N/A
2016/0189149	12/2015	MacLaurin et al.	N/A	N/A
2016/0192407	12/2015	Fyfe et al.	N/A	N/A
2016/0207454	12/2015	Cuddihy et al.	N/A	N/A
2016/0226313	12/2015	Okubo	N/A	N/A
2016/0236692	12/2015	Kleen et al.	N/A	N/A
2016/0292563	12/2015	Park	N/A	N/A
2016/0317060	12/2015	Connor	N/A	N/A
2016/0334901	12/2015	Rihn	N/A	N/A
2016/0336758	12/2015	Breedvelt-Schouten et al.	N/A	N/A
2016/0350581	12/2015	Manuel et al.	N/A	N/A
2016/0361032	12/2015	Carter et al.	N/A	N/A
2017/0010677	12/2016	Roh et al.	N/A	N/A
2017/0012925	12/2016	Tekin et al.	N/A	N/A
2017/0024008	12/2016	Kienzle et al.	N/A	N/A
2017/0026790	12/2016	Flitsch et al.	N/A	N/A
2017/0042477	12/2016	Haverinen et al.	N/A	N/A
2017/0053461	12/2016	Pal et al.	N/A	N/A
2017/0057492	12/2016	Eddington et al.	N/A	N/A
2017/0070078	12/2016	Hwang et al.	N/A	N/A
2017/0075701	12/2016	Ricci et al.	N/A	N/A
2017/0080952	12/2016	Gupta et al.	N/A	N/A
2017/0090475	12/2016	Choi et al.	N/A	N/A
2017/0109512	12/2016	Bower et al.	N/A	N/A
2017/0129335	12/2016	Lu et al.	N/A	N/A
2017/0131772	12/2016	Choi	N/A	N/A
2017/0190121	12/2016	Aggarwal et al.	N/A	N/A
2017/0192530	12/2016	Lee	N/A	N/A
2017/0242428	12/2016	Pal et al.	N/A	N/A
2017/0251967	12/2016	Premsukh	N/A	N/A
2017/0336964	12/2016	Kim	N/A	N/A
2017/0346635	12/2016	Gummeson et al.	N/A	N/A
2017/0347895	12/2016	Wei et al.	N/A	N/A
2017/0374074	12/2016	Stuntebeck	N/A	N/A
2018/0025351	12/2017	Chen et al.	N/A	N/A
2018/0025430	12/2017	Perl et al.	N/A	N/A
2018/0032126	12/2017	Liu	N/A	N/A
2018/0037228	12/2017	Biondo et al.	N/A	N/A
2018/0039303	12/2017	Hashimoto et al.	N/A	N/A
2018/0052428	12/2017	Abramov	N/A	N/A

2018/0054513	12/2017	Ma	N/A	N/A
2018/0068105	12/2017	Shapiro	N/A	G06F 21/602
2018/0093606	12/2017	Terwilliger et al.	N/A	N/A
2018/0093610	12/2017	Sun et al.	N/A	N/A
2018/0093672	12/2017	Terwilliger et al.	N/A	N/A
2018/0115797	12/2017	Wexler et al.	N/A	N/A
2018/0120891	12/2017	Eim	N/A	N/A
2018/0120892	12/2017	Von Badinski et al.	N/A	N/A
2018/0123629	12/2017	Wetzig	N/A	N/A
2018/0167200	12/2017	High et al.	N/A	N/A
2018/0174457	12/2017	Taylor	N/A	N/A
2018/0178712	12/2017	Terwilliger et al.	N/A	N/A
2018/0229674	12/2017	Heinrich et al.	N/A	N/A
2018/0256027	12/2017	Lacher	N/A	N/A
2018/0257668	12/2017	Tonshal	N/A	N/A
2018/0262505	12/2017	Ligatti	N/A	N/A
2018/0292901	12/2017	Priyantha et al.	N/A	N/A
2018/0300467	12/2017	Kwong et al.	N/A	N/A
2018/0322957	12/2017	Dill et al.	N/A	N/A
2019/0004325	12/2018	Connor	N/A	N/A
2019/0049267	12/2018	Huang	N/A	N/A
2019/0083022	12/2018	Huang	N/A	N/A
2019/0131812	12/2018	Lee et al.	N/A	N/A
2019/0155104	12/2018	Li et al.	N/A	N/A
2019/0155385	12/2018	Lim et al.	N/A	N/A
2019/0172289	12/2018	O'Toole et al.	N/A	N/A
2019/0191998	12/2018	Heikenfeld et al.	N/A	N/A
2019/0213429	12/2018	Sicconi et al.	N/A	N/A
2019/0230507	12/2018	Li et al.	N/A	N/A
2019/0265868	12/2018	Penilla et al.	N/A	N/A
2019/0286805	12/2018	Law et al.	N/A	N/A
2019/0287083	12/2018	Wurmfeld et al.	N/A	N/A
2019/0295440	12/2018	Hadad	N/A	N/A
2019/0298173	12/2018	Lawrence et al.	N/A	N/A
2019/0298265	12/2018	Keating	N/A	N/A
2019/0313967	12/2018	Lee	N/A	N/A
2019/0332140	12/2018	Wang et al.	N/A	N/A
2019/0342329	12/2018	Turgeman	N/A	N/A
2019/0357834	12/2018	Aarts et al.	N/A	N/A
2020/0001895	12/2019	Scheggi	N/A	N/A
2020/0005791	12/2019	Rakshit et al.	N/A	N/A
2020/0062276	12/2019	Yuan et al.	N/A	N/A
2020/0070840	12/2019	Gunaratne	N/A	N/A
2020/0142497	12/2019	Zhu	N/A	N/A
2020/0159896	12/2019	Shapiro et al.	N/A	N/A
2020/0218238	12/2019	Wang	N/A	N/A
2020/0356652	12/2019	Yamaguchi et al.	N/A	N/A
2020/0391696	12/2019	Kato et al.	N/A	N/A
2021/0019731	12/2020	Rule et al.	N/A	N/A
2021/0029112	12/2020	Palle et al.	N/A	N/A
2021/0058692	12/2020	Shankar	N/A	N/A
2021/0197849	12/2020	Tsuji	N/A	N/A
2021/0382684	12/2020	Hachiya et al.	N/A	N/A
2022/0083149	12/2021	Keller et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	**Application Date**	**Country**	**CPC**
103109462	12/2012	CN	N/A
104799509	12/2014	CN	N/A
105960196	12/2015	CN	N/A
106360895	12/2016	CN	N/A
206213423	12/2016	CN	N/A

206333477	12/2016	CN	N/A
206371611	12/2016	CN	N/A
107139933	12/2016	CN	N/A
107260139	12/2016	CN	N/A
104157116	12/2016	CN	N/A
105006103	12/2017	CN	N/A
105841851	12/2017	CN	N/A
108900691	12/2017	CN	N/A
108926081	12/2017	CN	N/A
10201301233399	12/2012	DE	N/A
102015006677	12/2015	DE	N/A
102019116618	12/2019	DE	N/A
1223191	12/2001	EP	N/A
1384752	12/2003	EP	N/A
2281205	12/2010	EP	N/A
2581856	12/2012	EP	N/A
200879676	12/2007	JP	N/A
20110012229	12/2010	KR	N/A
20170013067	12/2016	KR	N/A
1020170087113	12/2016	KR	N/A
101835991	12/2017	KR	N/A
2001017421	12/2000	WO	N/A
2005114476	12/2004	WO	N/A
2005124594	12/2004	WO	N/A
2008008714	12/2007	WO	N/A
2011132009	12/2010	WO	N/A
2015077418	12/2014	WO	N/A
2017136940	12/2016	WO	N/A
2018000396	12/2017	WO	N/A
2018048563	12/2017	WO	N/A
2018154341	12/2017	WO	N/A
2018164632	12/2017	WO	N/A
2018204811	12/2017	WO	N/A
2019082095	12/2018	WO	N/A
2019140528	12/2018	WO	N/A
2019180626	12/2018	WO	N/A

OTHER PUBLICATIONS

"How to find your ideal bedtime with the Oura app", available online at <https://web.archive.org/web/20191206205332/https://ouraring.com/how-find-your-ideal-bedtime-with-the-oura-app/>, 2019, 8 pages. 2019. cited by applicant

"Vauxhall/Opel In-Car Wireless Charging", retrieved from <https://www.air-charge.com/aircharge-for-business/automotive/vauxhall-wireless-charging>, Oct. 2019, 4 pages. 2019. cited by applicant

"Wireless charging for smart ring/pointing devices" available online at <http://www.humavox.com/smt_product/wireless-charging-for-smart-ringpointing-devices/>, Oct. 2019, 3 pages. 2019. cited by applicant

Adafruit, p. 1-2, available at: https://adafruit.com/product/2806, published Jun. 2019 (Year: 2019) 2019. cited by applicant

Adafruit.com, "RFID/NFC Smart Ring—Size 12—NTAG213," accessed at https://web.archive.org/web/20190605061438/https://www.adafruit.com/product/2806, publication Jun. 5, 2019 Jun. 5, 2019. cited by applicant

ASU projection wearable: Live tomorrow today (world first launch @ CES 2016). (Dec. 2015). ASU Tech, YouTube. Retrieved from https://www.youtube.com/watch?v=Wdb5O-D7Y0Y 2015. cited by applicant

Brownell, L., "Low-cost wearables manufactured by hybrid 3D printing. Wyss Institute, Harvard," Retrieved from https://wyss.harvard.edu/news/low-cost-wearables-manufactured-by-hybrid-3d-printing/, Sep. 6, 2017, pp. 11 2017. cited by applicant

Cetin, C., "Design, testing and implementation of a new authentication method using multiple devices," Graduate Theses and Dissertations, University of South Florida Scholar Commons. Retrieved from http://scholarcommons.usf.edu/etd/5660, Jan. 2015, pp. 61 Jan. 2015. cited by applicant

Charles Q. Choi, "Low Battery? New Tech Lets You Wirelessly Share Power", available online at <https://www.livescience.com/54790-new-tech-enables-wireless-charging.html>, May 19, 2016, 9 pages May 19, 2016. cited by applicant

Chen, X. A., et al., "Encore: 3D printed augmentation of everyday objects with printed-over, affixed and interlocked attachments," Nov. 5, 2015, pp. 73-82 2015. cited by applicant

Chen, X. A., et al., “Reprise: A design tool for specifying, generating, and customizing 3D printable adaptations on everyday objects,” Oct. 16, 2016, pp. 29-39 Oct. 16, 2016. cited by applicant
E-Senses, “Personal vitamin D, sunlight and daylight coach”, available online at <https://e-senses.com/>, 2019, 5 pages 2019. cited by applicant
Google translation of KR20170087113A (Year: 2016) 2016. cited by applicant
Hipolite, W., “The 3D printed O Bluetooth Ring is one of the tiniest personal computers you will ever see,” 3DPrint.com. Retrieved from https://3dprint.com/34627/o-bluetooth-ring-3d-printed/, Jan. 2015, pp. 5 Jan. 2015. cited by applicant
https://en.wikipdia.org/w/index.php?title=Ring_size&oldid=891328817 2019. cited by applicant
Hussain Almossawi, “This smart ring aims to provide better lives for people with sickle cell disease”, retrieved from <https://www.core77.com/projects/82131/This-Smart-Ring-Aims-to-Provide-Better-Lives-for-People-with-Sickle-Cell-Disease>, 2021, 9 pages. 2021. cited by applicant
Je et al., “PokeRing: Notifications by poking around the finger”, Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems—CHI'18, 2018, paper 542, pp. 1-10 2018. cited by applicant
Katharine Schwab, “Orii, the ring that turns your finger into a phone, is here”, available online at < https://www.fastcompany.com/90399237/orii-the-ring-that-turns-your-finger-into-a-phone-is-here >, 2019, 4 pages 2019. cited by applicant
Laput et al., “Skin buttons: cheap, small, low-powered and clickable fixed-icon laser projectors”, UIST '14:Proceedings of the 27th annual ACM symposium on User interface software and technology, Oct. 2014 pp. 389-394. 2014. cited by applicant
Magno et al., “Self-sustainable smart ring for long-term monitoring of blood oxygenation”, IEEE Access, 2019, pp. 115400-115408 2019. cited by applicant
Mahmud et al., Wearable technology for drug abuse detection: A survey of recent advancements, Smart Health, vol. 13, Aug. 2019, 100062 Aug. 2019. cited by applicant
Margaret, “The Orb: A Bluetooth headset that turns into a ring”, Gadgets, BornRich, Jun. 2013, available online at <http://www.bornrich.com/the-orb-a-bluetooth-headset-that-turns-into-a-ring.html >. Jun. 2013. cited by applicant
Mario, https://www.smartringnews.com/posts/smart-ring-vs-smartwatch-which-is-the-best-fitness-and-activity-tracker 2014. cited by applicant
Nassi et al., “Virtual breathalyzer”, Department of Software and Information Systems Engineering, Ben-Gurion University of the Negev, Israel, 2016, 10 pages 2016. cited by applicant
Neev Kiran, “SkinnySensor: Enabling Battery-Less Wearable Sensors via Intrabody Power Transfer”, Masters Theses 694, University of Massachusetts Amherst, 2018, 63 pages 2018. cited by applicant
Nerd-Fu, “Push present”, Delicious Juice Dot Com, Apr. 2015, available online at < https://blog.deliciousjuice.com/2015/04/ > Apr. 2015. cited by applicant
Pablo E SuArez, “NXT Ring—Your Digital-self at Hand”, available online at <https://www.youtube.com/watch?v=9w7uxDHs7NY>, uploaded on Jun. 21, 2019, 2 pages Jun. 21, 2019. cited by applicant
Roumen et al., “NotiRing: A comparative study of notification channels for wearable interactive rings”, Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems—CHI'15, 2015, pp. 2497-2500 2015. cited by applicant
Sarah Jacobsson Purewal, “Ringly review: The smart ring that could be sexier”, available online at <https://www.macworld.com/article/227133/ringly-review-the-smart-ring-that-could-be-sexier.html>, 2016, 10 pages 2016. cited by applicant
Schwab, K, “This startup wants to kill passwords—and replace them with jewelry. Fast Company, ” Retrieved from https://www.fastcompany.com/90254843/this-startup-wants-to-kill-passwords-and-replace-them-with-jewelry, (Oct. 2018), pp. 7 Oct. 2018. cited by applicant
Seung et al., “Nanopattemed Textile-Based Wearable Triboelectric Nanogenerator”, ACS Nano, vol. 9, 2015, pp. 3501-3509 2015. cited by applicant
Shane McGlaun, “Geek builds Bluetooth Smart Ring with OLED display”, available online at <https://www.slashgear.com/geek-builds-bluetooth-smart-ring-with-oled-display-02361383/>, 2015, 6 pages 2015. cited by applicant
Sperlazza, “We tested four sleep tracker apps and wearables: Here are the best ones”, available online at <https://www.bulletproof.com/sleep/tech/best-sleep-tracker-apps/>, 2019, 18 pages 2019. cited by applicant
Turunen, “Smart ring for stress control and self-understanding”, available online at <https://slowfinland.fi/en/smart-ring-for-stress-control-and-self-understanding/>, 2017, 9 pages 2017. cited by applicant
Woochit Tech (2017). New smart ring monitors UV exposure [video file]. Retrieved from https://www.youtube.com/watch?v=4YvkioTZxjU, 3 pages. 2017. cited by applicant
Worgan et al., “Garment level power distribution for wearables using inductive power transfer,” 9th International Conference on Human System Interactions (HSI), 2016, pp. 277-283 2016. cited by applicant
Xiao et al., “LumiWatch: On-arm projected graphics and touch input,” Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems—CHI'18, 2018, pp. 1-11 2018. cited by applicant
Zhu et al., “Developing a driving fatigue detection system using physiological sensors,” Proceedings of the 29th Australian Conference on Computer-Human Interaction—OZCHI'17, 2017, pp. 566-570 2017. cited by applicant
Zhu, M., et al., “Fluidic fabric muscle sheets for wearable and soft robotics,” Retrieved from https://arxiv.org/pdf/1903.08253.pdf Mar. 2019, pp. 32 Mar. 2019. cited by applicant
“The Oura App | Oura Ring”, available online at <https://web.archive.org/web/20191019192921/https://ouraring.com/introducing

the-new-oura-app/>, 6 pages Oct. 29, 2019. cited by applicant
"Get the Technical Specs of Oura Ring | Oura Ring", available online at <https://web.archive.org/web/2019129014439/https://ouraring.com/tech-specs/>, 3 pages Jan. 29, 2019. cited by applicant
"Learn how the Oura ring works | Go inside | Oura Ring", available online at <https://web.archive.org/web/20181127193557/https://ouraring.com/how-oura-works/>, 5 pages Nov. 27, 2018. cited by applicant
Oura Ring Generation 1 User Manual, available online at < https://fccid.io/2AD7V-OURARING15001/User-Manual/User-Manual-2844448.pdf>, 8 pages Nov. 24, 2015. cited by applicant
Oura Ring Generation 2 User Manual, available online at < https://fccid.io/2AD7V-OURA1801/User-Manual/User-manual-v2-3856414.pdf>, 13 pages Mar. 27, 2018. cited by applicant
Oura ring. Improve sleep. Perform better. by OURA—Kickstarter, available online at <https://web.archive.org/web/20160427015852/https://www.kickstarter.com/projects/oura/oura-ring-improve-sleep-perform-better/description>, 27 pages Apr. 27, 2016. cited by applicant
Oura Ring review—The Gadgeteer, available online at < https://the-gadgeteer.com/2017/08/11/oura-ring-review/> 16 pages Aug. 11, 2017. cited by applicant
Ouraring.com JZ50-0112 user manual available with Gen 2 ring purchase, 2 pages. cited by applicant
Important Information Please Read JZ50-0148 user manual available with gen 2 ring purchase, 4 pages May 13, 2019. cited by applicant
Important Information Please Read JZ50-0149 user manual available with gen 2 ring purchase, 4 pages May 13, 2019. cited by applicant
Oura Ring | Sleep Tracker and Smart Ring with a Heart Rate Monitor, available online at <https://web.archive.org/web/20180709050831/https://ouraring.com/>, 12 pages Jul. 9, 2018. cited by applicant
Introducing the New Oura Ring Generation 3—The Pulse Blog, available online at <https://ouraring.com/blog/oura-generation2-vs-generation3/>, 3 pages Oct. 26, 2021. cited by applicant
Oura Ring 2 Teardown: Inside the NBA's COVID-19-Detecting Smart Ring, available at < https://www.youtube.com/watch?v=BwA1hmSVgVY> (Screenshots) Jul. 9, 2020. cited by applicant
Oura Ring 2 Teardown: Inside the NBA's COVID-19-Detecting Smart Ring, available at < https://www.youtube.com/watch?v=BwA1hmSVgVY> (Script) Jul. 9, 2020. cited by applicant
Int'l Search Report and Written Opinion for PCT/IB2008/053982 Mar. 2009. cited by applicant
Hill et al., (2024) The Best Smart Ring to Rule Them All Retrieved from: https://www.wired.com/gallery/best-smart-rings/ 2024. cited by applicant
Kaltenbach, F. (2004). Translucent Materials: Glass, Plastics, Metals 2004. cited by applicant
International Search Report and Written Opinion—PCT/US2007/081119, International Search Authority—European Patent Office—Nov. 4, 2009. Nov. 2009. cited by applicant
Zhu et al., (2011). Wearable Sensor-Based Hand Gesture and Daily Activity Recognition for Robot-Assisted Living. IEEE Transactions on Systems, Man, and Cybernetics, Part A. 41. 569-573. 10.1109/TSMCA.2010.2093883. 2011. cited by applicant
Stables et al., (2022). Why the Oura Ring Gen 3 was our wearable of the year. Retrieved from: https://www.wareable.com/wearable-tech/why-the-oura-ring-3-was-our-wearable-of-the-year-2022 2022. cited by applicant
Sun et al., (2013). Wireless Power Transfer for Medical Microsystems. 10.1007/978-1-4614-7702-0. 2013. cited by applicant
U.S. Appl. No. 61/768,279 2013. cited by applicant
Sleep Lab validation of a wellness ring detecting sleep patterns based on photoplethysmogram, actigraphy and body temperature published Feb. 9, 2016, to Kinnunen (Year: 2016) 2016. cited by applicant
Rhee et al., (2000). Artifact-resistant, power-efficient design of finger-ring plethysmographic sensors. I. Design and analysis. 4. 2792-2795 vol. 4. 10.1109/IEMBS.2000.901443. 2000. cited by applicant
Rhee et al., (2000). Artifact-resistant, power-efficient design of finger-ring plethysmographic sensors. II. Prototyping and benchmarking. 4. 2796-2799 vol. 4. 10.1109/IEMBS.2000.901444. 2000. cited by applicant
Dynamic drinkware-type analysis for mestas. cited by applicant
Liu et al., (2009). UWave: Accelerometer-based personalized gesture recognition and its applications. Pervasive and Mobile Computing. 5. 657-675. 10.1016/j.pmcj.2009.07.007. 2009. cited by applicant
Trigueiros et al., (2019). A comparison of machine learning algorithms applied to hand gesture recognition. 2019. cited by applicant
Castaneda et al., (2018), Int J Biosens Biolectron. 2018, "A review on wearable photoplethysmography sensors and their potential future applications in health care"; 4(4):195-202. doi: 10.15406/ijbsbe.2018.04.00125. 2018. cited by applicant
Mendelson et al., (2006). A Wearable Reflectance Pulse Oximeter for Remote Physiological Monitoring. Conference proceedings . . . Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference. 1. 912-5. 10.1109/IEMBS.2006.260137. 2006. cited by applicant
Smiley, S., Active RFID vs. Passive RFID: What's the Difference? https://www.atlasrfidstore.com/rfid-insider/active-rfid-vs-passive-rfid/?srsltid=AfmBOoqhNhYwPPUSENIXB8LarZMm3TVQ4ugn4nTNUhfpy-9yYC_j0wdm Mar. 2016. cited by applicant
Lawton, G., (2022). Active vs. passive RFID tags: Which to choose: TechTarget. Retrieved from: https://www.techtarget.com/searcherp/tip/Active-vs-passive-RFID-tags-Which-to-choose Nov. 2022. cited by applicant
Amma et al., (2010). Airwriting recognition using wearable motion sensors. ACM International Conference Proceeding Series. 10

10.1145/1785455.1785465. 2010. cited by applicant
Zhou et al., (2012). Analysis and Selection of Features for Gesture Recognition Based on a Micro Wearable Device. International Journal of Advanced Computer Science and Applications. 3. 10.14569/IJACSA.2012.030101. 2012. cited by applicant
Rhee et al., (2001). Artifact-resistant power-efficient design of finger-ring plethysmographic sensors. IEEE transactions on bio-medical engineering. 48. 795-805. 10.1109/10.930904. 2001. cited by applicant
Ying et al., (2007). Automatic Step Detection in the Accelerometer Signal. 10.1007/978-3-540-70994-7_14. 2007. cited by applicant
100 Best Inventions of 2020 https://time.com/collection/best-inventions-2020/ 2020. cited by applicant
Clark, B. A.. “Color in Sunglass Lenses *. ” Optometry and Vision Science 46 (1969): 825-840. 1969. cited by applicant
Rhee, Sokwoo. (2006). Design and analysis of artifact-resistive finger photoplethysmographic sensors for vital sign monitoring /. 2006. cited by applicant
Webster, J.G., Design of pulse oximeters Webster—Institute of Physics Publishing—2003 2003. cited by applicant
Park et al., (2011). E-gesture: A collaborative architecture for energy-efficient gesture recognition with hand-worn sensor and mobile devices. SenSys 2011—Proceedings of the 9th ACM Conference on Embedded Networked Sensor Systems. 359-360. 10.1145/1999995.2000034. 2011. cited by applicant
Teh et al., (2000). Embedding of electronics within thermoplastic polymers using injection moulding technique. Filtration Industry Analyst. 10-18. 10.1109/IEMT.2000.910703. 2000. cited by applicant
Sakai, Tadamoto. (1993). Encapsulation process for electronic devices using injection molding method. Advances in Polymer Technology. 12. 61-71. 10.1002/adv.1993.060120106. 1993. cited by applicant
Ardebili et al.—William Andrew—2019. Encapsulation technologies for electronic applications 2019. cited by applicant
Au et al., (2009). Episodic Sampling: Towards Energy-efficient Patient Monitoring with Wearable Sensors. Conference proceedings : . . . Annual International Conference of the IEEE Engineering in Medicine and Biology Society. IEEE Engineering in Medicine and Biology Society. Conference. 2009. 6901-5. 10.1109/IEMBS.2009.5333615. 2009. cited by applicant
Silverman, A. (2002). Fifty Years of Glass-Making. Industrial & Engineering Chemistry. 18. 10.1021/ie50201a004. 2002. cited by applicant
Petropoulos et al., (2012). Flexible PCB-MEMS flow sensor. Procedia Engineering. 47. 236-239. 10.1016/j.proeng.2012.09.127. 2012. cited by applicant
Schlömer et al., (2008). Gesture Recognition with a Wii Controller. First publ. in: Proceedings of the 2nd International Conference on Tangible and Embedded Interaction 2008, Bonn, Germany, Feb. 18-20, 2008, pp. 11-14. 10.1145/1347390.1347395. 2008. cited by applicant
Rekimoto, J. (2001). GestureWrist and GesturePad: unobtrusive wearable interactiondevices. International Symposium on Wearable Computers, Digest of Papers. 21-27. 10.1109/ISWC.2001.962092. 2001. cited by applicant
Guidelines to Enhancing the Heart-Rate Monitoring Performance of Biosensing Wearables https://www.analog.com/en/resources/technical-articles/guidelines-to-enhancing-the-heartrate-monitoring-performance-of-biosensing-wearables.html 2019. cited by applicant
Krzyanowski, J. “How are PCB's made? A Beginner's Guide to the PCB Manufacturing Process” Retrieved from Knowledge zone https://vectorbluehub.com/how-are-pcbs-made 2023. cited by applicant
Ciofu, C., et al., (2013) Injection and Micro Injection of Polymeric . . . First edition of the International Scientific Conference Modern Technologies in Machine Manufacturing Technology TMCM ISSN 2067-3604, vol. V, No. 1/2013https://modtech.ro/international-journal/vol5no12013/Ciofu_Ciprian_1.pdf 2013. cited by applicant
Ross, R.J.. (2004). LCP injection molded packages—keys to JEDEC 1 performance. 1807-1811 vol. 2. 10.1109/ECTC.2004.1320364. 2004. cited by applicant
Murphy, K. (2012) Machine learning: a probabilistic perspective 2012. cited by applicant
Olofson et al., Machining of titanium alloys—Battelle Memorial Institute, Defense Metals Information Center—1965 1965. cited by applicant
Asada et al., “Mobile monitoring with wearable photoplethysmographic biosensors,” in IEEE Engineering in Medicine and Biology Magazine, vol. 22, No. 3, pp. 28-40, May-Jun. 2003, doi: 10.1109/MEMB.2003.1213624 2003. cited by applicant
Chen et al., (2009). Monitoring Human Movements at Home Using Wearable Wireless Sensors. Engineering Faculty Presentations. 2009. cited by applicant
On the Heels of 1 Million Rings Sold, Oura Now Valued at $2.55 Billion https://www.businesswire.com/news/home/20220405006108/en/On-the-Heels-of-1-Million-Rings-Sold-Oura-Now-Valued-at-2.55-Billion#:˜:text=SAN%20FRANCISCO%2D%2D(BUSINESS%20WIRE,of%20selling%201%2C000%2C000%20Oura%20Rings Apr. 2022. cited by applicant
Lister et al., (2018) Optical properties of human skin https://www.spiedigitallibrary.org/journals/journal-of-biomedical-optics/volume17/issue-9/090901/Optical-properties-of-human-skin/10.1117/1.JBO.17.9.090901.full 2018. cited by applicant
Uchino et al., (2000). Prediction of Optical Properties of Commercial Soda-Lime-Silicate Glasses Containing Iron. Journal of Non-Crystalline Solids. 261. 72-78. 10.1016/S0022-3093(99)00617-1. 2000. cited by applicant
Lawrence et al., (1989) “Pulse Oximetry” Anesthesiology: The journal of the American Society of Anesthesiologists, Inc. vol. 70 No. 1 1989. cited by applicant

Konig et al., (1998) “Reflectance Pulse Oximetry-Principles and obstetric application in the zurich system” Journal of Clinical Monitoring and Computing vol. 14 No. 6PG. 3-6 1998. cited by applicant

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS (1) This application is a continuation of U.S. Non-Provisional patent application Ser. No. 18/187,745, filed on Mar. 22, 2023, which claims priority to U.S. Non-Provisional Patent application Ser. No. 16/926,270 filed on Jul. 10, 2020, which claims priority to U.S. Provisional Patent Application No. 62/877,391, filed Jul. 23, 2019, and U.S. Provisional Patent Application No. 63/004,789, filed Apr. 3, 2020, all of which are incorporated by reference herein for all purposes.

FIELD OF THE DISCLOSURE
(1) The present disclosure generally relates to implementations of smart ring wearable devices and, more specifically, to charging smart ring devices using energy harvested from motion.
BACKGROUND
(2) To the extent that smart ring technology has been adopted, it has a number of challenges. For example, a number of problems exist with wearable devices generally, including: they often need to be removed for charging; they often have poor fit; they often provide relatively little user interactivity; and they often provide limited functionality.
BRIEF SUMMARY
(3) A smart ring may be configured to harvest energy from user motion using a piezoelectric effect. To that end, the smart ring may include a piezoelectric harvesting element configured to generate a voltage in response to a deformation caused by motion.
(4) In one aspect, a smart ring includes a ring-shaped housing, a power source disposed within the ring-shaped housing, and a charging circuit. The charging circuit includes a piezoelectric harvesting element, and is configured to charge the power source by way of electrical energy generated by mechanical deformation within the piezo-electric harvesting element. The smart ring further includes a component, disposed within the ring-shaped housing and configured to draw energy from the power source, and further configured to perform at least one of: i) sense a physical phenomenon external to the ring-shaped housing, ii) send communication signals to a communication device external to the ring-shaped housing, or iii) implement a user interface.
(5) In another aspect, a method for operating a smart ring includes generating electrical energy at a charging circuit within the smart ring by way of a mechanical deformation in a piezoelectric harvesting element of the charging circuit. The method further includes charging a power source disposed within the ring-shaped housing of the smart ring by way of the charging circuit configured to deliver the generated electrical energy to the power source. Still further, the method includes operating a component, disposed within the ring-shaped housing of the smart ring and configured to draw energy from the power source, and further configured to perform at least one of: i) sense a physical phenomenon external to the ring-shaped housing, ii) send communication signals to a communication device external to the ring-shaped housing, or iii) implement a user interface.
(6) In another embodiment, a smart ring includes a housing configured to be worn by a user. The smart ring can further include a power source disposed within the housing. The smart ring also can include a charging circuit disposed within the housing. The charging circuit can include a harvesting element configured to charge the power source. The harvesting element can be configured to convert mechanical energy to electrical energy. The smart ring also can include a device at least partially disposed within the housing. The device can be configured to draw energy from the power source.
(7) In another embodiment, a smart ring includes a means for housing configured to be worn by a user. The smart ring can further include a means for powering disposed within the means for housing. The smart ring also can include a means for charging disposed within the means for housing. The means for charging can include a means for harvesting configured to charge the means for powering. The means for harvesting can be configured to convert mechanical energy to electrical energy. The smart ring also can include a means for drawing energy from the means for powering. The means for drawing energy from the means for powering can be at least partially disposed within the means for housing.
(8) In another embodiment, a method for operating a smart ring includes generating electrical energy at a charging circuit within a smart ring. The charging circuit can include a harvesting element. The harvesting element can be configured to convert mechanical energy to electrical energy. The method for operating a smart ring further can include charging a power source disposed within the smart ring by delivering the electrical energy from the charging circuit to the power source. The method for operating a smart ring also can include causing the power source to provide power to a device disposed within the smart ring.
(9) Depending upon the embodiment, one or more benefits may be achieved. These benefits and various additional

objects, features and advantages of the present disclosure can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** illustrates a system comprising a smart ring and a block diagram of smart ring components, according to some embodiments.

(2) FIG. **2** illustrates a number of different form factor types of a smart ring, according to some embodiments.

(3) FIG. **3** illustrates examples of different smart ring form factors.

(4) FIG. **4** illustrates an environment within which a smart ring may operate, according to some embodiments.

(5) FIG. **5** is a schematic diagram of a charging system for a smart ring, according to some embodiments.

(6) FIG. **6**A illustrates an example charging system that includes one or more permanent magnets disposed near or attached to a keyboard.

(7) FIG. **6**B illustrates a configuration of an induction coil for a charging system illustrated in FIG. **6**A, according to some embodiments.

(8) FIG. **7**A illustrates an example charging system integrated into a smart ring that includes one or more permanent magnets disposed within a band of the smart ring.

(9) FIG. **7**B illustrates a section of the smart ring in FIG. **7**A, according to some embodiments.

(10) FIG. **8** is a schematic diagram of a smart ring configured to recharge by way of a piezoelectric harvesting element according to some embodiments.

(11) FIG. **9**A illustrates a smart ring configuration with two bands configured to move with respect to one another to generate mechanical energy for piezoelectric harvesting, according to some embodiments.

(12) FIG. **9**B illustrates a mechanism for causing deformation in a piezoelectric harvesting element when the bands in FIG. **9**A move with respect to one another, according to some embodiments.

DETAILED DESCRIPTION

(13) Smart ring wearable technology can enable a wide range of applications including security, safety, health and wellness, and convenient interfacing between a user and a variety of technologies based at least in part upon integrating a variety of sensor, input/output devices, and computing capabilities in a compact form factor. One of the challenges in increasing smart ring capabilities is reliably powering the needed components, particularly considering the limited space for a power source in the compact form factor. An ability to conveniently charge the power source of the smart ring without removing the smart ring from a finger would contribute to the adoption of smart ring technology.

(14) One way to charge the smart ring without removing from the finger may include using a wearable charger, or a charger disposed in the environment of the smart ring. For the purposes of charging, the charger may be connected to the smart ring by a cable. Additionally or alternatively, the smart ring may include one or more circuits that enable the smart ring to harvest optical, thermal, mechanical and/or other sources of energy to charge and/or recharge the smart ring. Some techniques for harvesting mechanical energy may be based at least in part upon electromagnetic induction, or a piezoelectric effect, as described herein. Some techniques may rely on generated variable electromagnetic fields at a charging source that may couple a portion of the generated energy to a smart ring. Such configurations may be referred to as wireless charging, rather than harvesting configurations. A charging system may generate constant magnetic fields with constant current through a properly configured conductor. Such systems rely on availability of electrical energy to sustain a magnetic field. An electromagnetic induction charging system that uses permanent magnets has the advantage of obviating the need for using electrical energy to produce magnetic fields, on the other hand.

(15) Various techniques, systems, and methods for charging a power source of a smart ring using electromagnetic induction are discussed below with reference to FIG. **1**, FIG. **2**, FIG. **3**, FIG. **4**, FIG. **5**, FIG. **6**A, FIG. **6**B, FIG. **7**A, FIG. **7**B, FIG. **8**, FIG. **9**A, and FIG. **9**B. In section I, a smart ring and inductive charging system is described with reference to FIG. **1**. In section II, example smart ring form factor types and configurations to facilitate harvesting energy via induction are discussed with reference to FIG. **2** and FIG. **3**. In section III, an example operating environment in which a smart ring may operate is described with reference to FIG. **4**. In section IV, an example schematic for an induction charging system is described with reference to FIG. **5**. In section V, example induction charging systems are described with reference to FIG. **6**A, FIG. **6**B, FIG. **7**A, and FIG. **7**B. In section VI, a smart ring configured for charging using a piezoelectric harvesting element is described with reference to FIG. **8**, FIG. **9**A, and FIG. **9**B. In section VII, other considerations are described.

I. Examples of Smart Ring and Smart Ring Components

(16) FIG. **1** illustrates a system **100** comprising a smart ring **101** that may be charged via an induction charging system (e.g., by way of harvesting motion energy) according to one or more of the techniques described herein. FIG. **1** also shows one or more devices or systems that may be electrically, mechanically, or communicatively connected to the smart ring **101**. As shown, the smart ring **101** may include a set of components **102**, which may have various power needs and may impact the frequency with which the smart ring **101** needs recharging. To facilitate compatibility with the described energy harvesting techniques, the components **102** may be configured to be compatible with the magnetic fields and magnetic field variations to which the smart ring **101** and the components **102** may be exposed during induction charging. With respect to the described piezoelectric harvesting, the components **102** may be configured to be isolated from the

potentially high voltages generated during piezoelectric harvesting.
(17) The system **100** may comprise any one or more of: a charger **103** for the smart ring **101**, a user device **104**, a network **105**, a mobile device **106**, or a server **107**. The charger **103** may provide energy to the smart ring **101** by way of a direct electrical, a wireless, or an optical connection. The smart ring **101** may be in a direct communicative connection with the user device **104**, the mobile device **106**, or the server **107** by way of the network **105**. Interactions between the smart ring **101** and other components of the system **100** are discussed in more detail in the context of FIG. **4**.
(18) The smart ring **101** may sense a variety of signals indicative of activities of a user wearing the ring **101**, biometric signals, a physiological state of the user, or signals indicative of the user's environment. The smart ring **101** may analyze the sensed signals using built-in computing capabilities or in cooperation with other computing devices (e.g., user device **104**, mobile device **106**, server **107**) and provide feedback to the user or about the user via the smart ring **101** or other devices (e.g., user device **104**, mobile device **106**, server **107**). Additionally or alternatively, the smart ring **101** may provide the user with notifications sent by other devices, enable secure access to locations or information, or a variety of other applications pertaining to health, wellness, productivity, or entertainment.
(19) The smart ring **101**, which may be referred to herein as the ring **101**, may comprise a variety of mechanical, electrical, optical, or any other suitable subsystems, devices, components, or parts disposed within, at, throughout, or in mechanical connection to a housing **110** (which may be ring shaped and generally configured to be worn on a finger). Additionally, a set of interface components **112***a* and **112***b* may be disposed at the housing, and, in particular, through the surface of the housing. The interface components **112***a* and **112***b* may provide a physical access (e.g., electrical, fluidic, mechanical, or optical) to the components disposed within the housing. The interface components **112***a* and **112***b* may exemplify surface elements disposed at the housing. As discussed below, some of the surface elements of the housing may also be parts of the smart ring components.
(20) As shown in FIG. **1**, the components **102** of the smart ring **101** may be distributed within, throughout, or on the housing **110**. As discussed in the contexts of FIG. **2** and FIG. **3** below, the housing **110** may be configured in a variety of ways and include multiple parts. The smart ring components **102** may for example, be distributed among the different parts of the housing **110**, as described below, and may include surface elements of the housing **110**. The housing **110** may include mechanical, electrical, optical, or any other suitable subsystems, devices, components, or parts disposed within or in mechanical connection to the housing **110**, including a battery **120**, a charging unit **130**, a controller **140**, a sensor system **150** comprising one or more sensors, a communications unit **160**, a one or more user input devices **170**, or a one or more output devices **190**. Each of the components **120**, **130**, **140**, **150**, **160**, **170**, **190** may include one or more associated circuits, as well as packaging elements. The components **120**, **130**, **140**, **150**, **160**, **170**, **190** may be electrically or communicatively connected with each other (e.g., via one or more busses or links, power lines, etc.), and may cooperate to enable “smart” functionality described within this disclosure.
(21) The battery **120** may supply energy or power to the controller **140**, the sensors **150**, the communications unit **160**, the user input devices **170**, or the output devices **190**. In some scenarios or implementations, the battery **120** may supply energy or power to the charging unit **130**. The charging unit **130**, may supply energy or power to the battery **120**. In some implementations, the charging unit **130** may supply (e.g., from the charger **103**, or harvested from other sources) energy or power to the controller **140**, the sensors **150**, the communications unit **160**, the user input devices **170**, or the output devices **190**. In a charging mode of operation of the smart ring **101**, the average power supplied by the charging unit **130** to the battery **120** may exceed the average power supplied by the battery **120** to the charging unit **130**, resulting in a net transfer of energy from the charging unit **130** to the battery **120**. In a non-charging mode of operation, the charging unit **130** may on average, draw energy from the battery **120**.
(22) The battery **120** may include one or more cells that convert chemical, thermal, nuclear or another suitable form of energy into electrical energy to power other components or subsystems **140**, **150**, **160**, **170**, **190** of the smart ring **101**. The battery **120** may include one or more alkaline, lithium, lithium-ion and or other suitable cells. The battery **120** may include two terminals that, in operation, maintain a substantially fixed voltage of 1.5, 3, 4.5, 6, 9, 12 V or any other suitable terminal voltage between them. When fully charged, the battery **120** may be capable of delivering to power-sinking components an amount of charge, referred to herein as “full charge,” without recharging. The full charge of the battery may be 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000, 5000 mAh or any other suitable charge that can be delivered to one or more power-consuming loads as electrical current.
(23) The battery **120** may include a charge-storage device, such as, for example a capacitor or a super-capacitor. In some implementations discussed below, the battery **120** may be entirely composed of one or more capacitive or charge-storage elements. The charge storage device may be capable of delivering higher currents than the energy-conversion cells included in the battery **120**. Furthermore, the charge storage device may maintain voltage available to the components or subsystems **130**, **140**, **150**, **160**, **170**, **190** when one or more cells of the battery **120** are removed to be subsequently replaced by other cells.
(24) The charging unit **130** may be configured to replenish the charge supplied by the battery **120** to power-sinking components or subsystems (e.g., one or more of subsystems **130**, **140**, **150**, **160**, **170**, **190**) or, more specifically, by their associated circuits. To replenish the battery charge, the charging unit **130** may convert one form of electrical energy into another form of electrical energy. More specifically, the charging unit **130** may convert alternating current (AC) to direct current (DC), may perform frequency conversions of current or voltage waveforms, or may convert energy stored in static electric fields or static magnetic fields into direct current. Additionally or alternatively, the charging unit **130** may harvest

energy from radiating or evanescent electromagnetic fields (including optical radiation) and convert it into the charge stored in the battery **120**. Furthermore, the charging unit **130** may convert non-electrical energy into electrical energy. For example, the charging unit **130** may harvest energy from motion, or from thermal gradients.

(25) The controller **140** may include a processor unit **142** and a memory unit **144**. The processor unit **142** may include one or more processors, such as a microprocessor (μP), a digital signal processor (DSP), a central processing unit (CPU), a graphical processing unit (GPU), a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), or any other suitable electronic processing components. Additionally or alternatively, the processor unit **142** may include photonic processing components.

(26) The memory unit **144** may include one or more computer memory devices or components, such as one or more registers, RAM, ROM, EEPROM, or on-board flash memory. The memory unit **144** may use magnetic, optical, electronic, spintronic, or any other suitable storage technology. In some implementations, at least some of the functionality the memory unit **144** may be integrated in an ASIC or and FPGA. Furthermore, the memory unit **144** may be integrated into the same chip as the processor unit **142** and the chip, in some implementations, may be an ASIC or an FPGA.

(27) The memory unit **144** may store a smart ring (SR) routine **146** with a set of instructions, that, when executed by the processor **142** may enable the operation and the functionality described in more detail below. Furthermore, the memory unit **144** may store smart ring (SR) data **148**, which may include (i) input data used by one or more of the components **102** (e.g., by the controller when implementing the SR routine **146**) or (ii) output data generated by one or more of the components **102** (e.g., the controller **140**, the sensor unit **150**, the communication unit **160**, or the user input unit **170**). In some implementations, other units, components, or devices may generate data (e.g., diagnostic data) for storing in the memory unit **144**.

(28) The processing unit **142** may draw power from the battery **120** (or directly from the charging unit **130**) to read from the memory unit **144** and to execute instructions contained in the smart ring routine **146**. Likewise, the memory unit **144** may draw power from the battery **120** (or directly from the charging unit **130**) to maintain the stored data or to enable reading or writing data into the memory unit **144**. The processor unit **142**, the memory unit **144**, or the controller **140** as a whole may be capable of operating in one or more low-power mode. One such low power mode may maintain the machine state of the controller **140** when less than a threshold power is available from the battery **120** or during a charging operation in which one or more battery cells are exchanged.

(29) The controller **140** may receive and process data from the sensors **150**, the communications unit **160**, or the user input devices **170**. The controller **140** may perform computations to generate new data, signals, or information. The controller **140** may send data from the memory unit **144** or the generated data to the communication unit **160** or the output devices **190**. The electrical signals or waveforms generated by the controller **140** may include digital or analog signals or waveforms. The controller **140** may include electrical or electronic circuits for detecting, transforming (e.g., linearly or non-linearly filtering, amplifying, attenuating), or converting (e.g., digital to analog, analog to digital, rectifying, changing frequency) of analog or digital electrical signals or waveforms.

(30) The sensor unit **150** may include one or more sensors disposed within or throughout the housing **110** of the ring **101**. Each of the one or more sensors may transduce one or more of: light, sound, acceleration, translational or rotational movement, strain, temperature, chemical composition, surface conductivity or other suitable signals into electrical or electronic sensors or signals. A sensor may be acoustic, photonic, micro-electro-mechanical systems (MEMS) sensors, chemical, micro-fluidic (e.g., flow sensor), or any other suitable type of sensor. The sensor unit **150** may include, for example, an inertial motion unit (IMU) for detecting orientation and movement of the ring **101**.

(31) The communication unit **160** may facilitate wired or wireless communication between the ring **101** and one or more other devices. The communication unit **160** may include, for example, a network adaptor to connect to a computer network, and, via the network, to network-connected devices. The computer network may be the Internet or another type of suitable network (e.g., a personal area network (PAN), a local area network (LAN), a metropolitan area network (MAN), a wide area network (WAN), a mobile, a wired or wireless network, a private network, a virtual private network, etc.). The communication unit **160** may use one or more wireless protocols, standards, or technologies for communication, such as Wi-Fi, near field communication (NFC), Bluetooth, or Bluetooth low energy (BLE). Additionally or alternatively, the communication unit **160** may enable free-space optical or acoustic links. In some implementations, the communication unit **160** may include one or more ports for a wired communication connections. The wired connections used by the wireless communication module **160** may include electrical or optical connections (e.g., fiber-optic, twisted-pair, coaxial cable).

(32) User input unit **170** may collect information from a person wearing the ring **101** or another user, capable of interacting with the ring **101**. In some implementations, one or more of the sensors in the sensor unit **150** may act as user input devices within the user input unit **170**. User input devices may transduce tactile, acoustic, video, gesture, or any other suitable user input into digital or analog electrical signal, and send these electrical signals to the controller **140**.

(33) The output unit **190** may include one or more devices to output information to a user of the ring **101**. The one or more output devices may include acoustic devices (e.g., speaker, ultrasonic); haptic (thermal, electrical) devices; electronic displays for optical output, such as an organic light emitting device (OLED) display, a laser unit, a high-power light-emitting device (LED), etc.; or any other suitable types of devices. For example, the output unit **190** may include a projector that projects an image onto a suitable surface. In some implementations, the sensor unit **150**, the user input unit **170**, and the output unit **190** may cooperate to create a user interface with capabilities (e.g., a keyboard) of much larger

computer systems, as described in more detail below.

(34) The components **120**, **130**, **140**, **150**, **160**, **170**, **190** may be interconnected by a bus **195**, which may be implemented using one or more circuit board traces, wires, or other electrical, optoelectronic, or optical connections. The bus **195** may be a collection of electrical power or communicative interconnections. The communicative interconnections may be configured to carry signals that conform to any one or more of a variety of protocols, such as I2C, SPI, or other logic to enable cooperation of the various components.

II. Example Smart Ring Form Factor Types

(35) FIG. **2** includes block diagrams of a number of different example form factor types or configurations **205***a*, **205***b*, **205***c* of a smart ring (e.g., the smart ring **101**). When the smart ring uses electromagnetic induction charging using permanent magnets, a form factor and configuration of the smart ring may influence or, conversely, depend on the implementation of a charging circuit. In particular, the configuration of an induction coil may be tied to a form factor of the smart ring, as discussed below.

(36) The configurations **205***a*, **205***b*, **205***c* (which may also be referred to as the smart rings **205***a*, **205***b*, **205***c*) may each represent an implementation of the smart ring **101**, and each may include any one or more of the components **102** (or components similar to the components **102**). In some embodiments, one or more of the components **102** may not be included in the configurations **205***a*, **205***b*, **205***c*. The configurations **205***a*, **205***b*, **205***c* include housings **210***a*, **210***b*, **210***c*, which may be similar to the housing **110** shown in FIG. **1**.

(37) The configuration **205***a* may be referred to as a band-only configuration comprising a housing **210***a*. In the configuration **205***b*, a band may include two or more removably connected parts, such as the housing parts **210***b* and **210***c*. The two housing parts **210***b* and **210***c* may each house at least some of the components **102**, distributed between the housing parks **210***b* and **210***c* in any suitable manner.

(38) The configuration **205***c* may be referred to as a band-and-platform configuration comprising (i) a housing component **210***d* and (ii) a housing component **210***e* (sometimes called the “platform **210***e*”), which may be in a fixed or removable mechanical connection with the housing **210***d*. The platform **210***e* may function as a mount for a “jewel” or for any other suitable attachment. The housing component **210***d* and the platform **210***e* may each house at least one or more of the components **102** (or similar components).

(39) In some instances, the term “smart ring” may refer to a partial ring that houses one or more components (e.g., components **102**) that enable the smart ring functionality described herein. The configurations **205***d* and **205***e* may be characterized as “partial” smart rings, and may be configured for attachment to a second ring. The second ring may be a conventional ring without smart functionality, or may be second smart ring, wherein some smart functionality of the first or second rings may be enhanced by the attachment.

(40) The configuration **205***d*, for example, may include a housing **210***f* with a groove to enable clipping onto a conventional ring. The grooved clip-on housing **210***f* may house the smart ring components described above. The configuration **205***e* may clip onto a conventional ring using a substantially flat clip **210***g* part of the housing and contain the smart ring components in a platform **210***h* part of the housing.

(41) The configuration **205***f*, on the other hand, may be configured to be capable of being mounted onto a finger of a user without additional support (e.g., another ring). To that end, the housing **210***i* of the configuration **205***f* may be substantially of a partial annular shape subtending between 180 and 360 degrees of a full circumference. When implemented as a partial annular shape, the housing **210***i* may be more adaptable to fingers of different sizes that a fully annular band (360 degrees), and may be elastic. A restorative force produced by a deformation of the housing **210***i* may ensure a suitable physical contact with the finger. Additional suitable combinations of configurations (not illustrated) may combine at least some of the housing features discussed above.

(42) The configuration **205***g* may be configured to have two rings, a first ring **205***g***1** capable of and adapted to be mounted onto a finger of a user, and a second ring **205***g***2** capable of and adapted to be directly mounted onto the first ring **205***g***1**, as depicted in FIG. **2**. Said another way, the first ring **205***g***1** and the second ring **205***g***2** are arranged in a concentric circle arrangement, such that the second ring **205***g***2** does not contact a user's finger when the smart ring **205***g* is worn. Rather, only the first ring **205***g***1** contacts the user's finger. Additional suitable combinations of configurations (not illustrated) may combine at least some of the housing features discussed above.

(43) FIG. **3** includes perspective views of example configurations **305***a*, **305***b*, **305***c* of a smart right (e.g., the smart ring **101**) in which a number of surface elements are included. When the smart ring uses electromagnetic induction charging using permanent magnets, a form factor and configuration of the smart ring may influence or, conversely, depend on the implementation of a charging circuit. In particular, the configuration of an induction coil may be tied to a form factor of the smart ring, as discussed below. Additionally, some of the surface elements may be configured to indicate when a magnetic field at the smart ring rises above or falls below a threshold.

(44) Configuration **305***a* is an example band configuration **205***a* of a smart ring (e.g., smart ring **101**). Some of the surface elements of the housing may include interfaces **312***a*, **312***b* that may be electrically connected to, for example, the charging unit **130** or the communications unit **160**. On the outside of the configuration **305***a*, the interfaces **312***a*, **312***b* may be electrically or optically connected with a charger to transfer energy from the charger to a battery (e.g., the battery **120**), or with another device to transfer data to or from the ring **305***a*. The outer surface of the configuration **305***a* may include a display **390***a*, while the inner surface may include a biometric sensor **350***a*.

(45) The configurations **305***b* and **305***c* are examples of configurations of a smart ring with multiple housing parts (e.g.,

configuration **205***b* in FIG. **2**). Two (or more) parts may be separate axially (configuration **305***b*), azimuthally (configuration **305***c*), or radially (nested rings, not shown). The parts may be connected mechanically, electrically, or optically via, for example, interfaces analogous to interfaces **312***a*, **312***b* in configuration **305***a*. Each part of a smart ring housing may have one or more surface elements, such as, for example, sensors **350***b*, **350***c* or output elements **390***b*, **390***c*. The latter may be LEDs (e.g., output element **390***b*) or haptic feedback devices (e.g., output element **390***c*), among other suitable sensor or output devices. Additionally or alternatively, at least some of the surface elements (e.g., microphones, touch sensors) may belong to the user input unit **170**.

(46) Configuration **305***d* may be an example of a band and platform configuration (e.g., configuration **205***c*), while configurations **305***e* and **305***f* may be examples of the partial ring configurations **205***d* and **205***e*, respectively. Output devices **390***d*, **390***e*, **390***f* on the corresponding configurations **305***d*, **305***e*, **305***f* may be LCD display, OLED displays, e-ink displays, one or more LED pixels, speakers, or any other suitable output devices that may be a part of a suite of outputs represented by an output unit (e.g., output unit **190**). Other surface elements, such as an interface component **312***c* may be disposed within, at, or through the housing. It should be appreciated that a variety of suitable surface elements may be disposed at the illustrated configurations **305***a*, **305***b*, **305***c* at largely interchangeable locations. For example, the output elements **390***d*, **390***e*, **390***f* may be replaced with sensors (e.g., UV sensor, ambient light or noise sensors, etc.), user input devices (e.g., buttons, microphones, etc.), interfaces (e.g., including patch antennas or optoelectronic components communicatively connected to communications units), or other suitable surface elements.

III. Example Operating Environments for a Smart Ring

(47) FIG. **4** illustrates an example environment **400** within which a smart ring **405** may be configured to operate. A portion of an electromagnetic induction charging system (e.g., one or more permanent magnets) may be disposed within the environment, as discussed below. In an embodiment, the smart ring **405** may be the smart ring **101**. In some embodiments, the smart ring **405** may be any suitable smart ring capable of providing at least some of the functionality described herein. Depending on the embodiment, the smart ring **405** may be configured in a manner similar or equivalent to any of the configurations **205***a*, **205***b*, **205***c* or **305***a*, **305***b*, **305***c* shown in FIG. **2** and FIG. **3**.

(48) The smart ring **405** may interact (e.g., by sensing, sending data, receiving data, receiving energy) with a variety of devices, such as bracelet **420** or another suitable wearable device, a mobile device **422** (e.g., a smart phone, a tablet, etc.) that may be, for example, the user device **104**, another ring **424** (e.g., another smart ring, a charger for the smart ring **405**, etc.), a secure access panel **432**, a golf club **434** (or another recreational accessory), a smart ring **436** worn by another user, or a steering wheel **438** (or another vehicle interface). Additionally or alternatively, the smart ring **405** may be communicatively connected to a network **440** (e.g., WiFi, 5G cellular), and by way of the network **440** (e.g., network **105** in FIG. **1**) to a server **442** (e.g., server **107** in FIG. **1**) or a personal computer **444** (e.g., mobile device **106**). Additionally or alternatively, the ring **405** may be configured to sense or harvest energy from natural environment, such as the sun **450**.

(49) The ring **405** may exchange data with other devices by communicatively connecting to the other devices using, for example, the communication unit **160**. The communicative connection to other device may be initiated by the ring **405** in response to user input via the user input unit **170**, in response to detecting trigger conditions using the sensor unit **150**, or may be initiated by the other devices. The communicative connection may be wireless, wired electrical connection, or optical. In some implementation, establishing a communicative link may include establishing a mechanical connection.

(50) The ring **405** may connect to other devices (e.g., a device with the charger **103** built in) to charge the battery **120**. The connection to other devices for charging may enable the ring **405** to be recharged without the need for removing the ring **405** from the finger. For example, the bracelet **420** may include an energy source that may transfer the energy from the energy source to the battery **120** of the ring **405** via the charging unit **430**. To that end, an electrical (or optical) cable may extend from the bracelet **420** to an interface (e.g., interfaces **112***a*, **112***b*, **312***a*, **312***b*) disposed at the housing (e.g., housings **110**, **210***a*, **210***b*, **210***c*, **210***d*, **210***e*, **210***f*, **210***g*, **210***h*, **210***i*) of the ring **405**. The mobile device **422**, the ring **424**, the golf club **434**, the steering wheel **438** may also include energy source configured as chargers (e.g., the charger **103**) for the ring **405**. The chargers for may transfer energy to the ring **405** via a wired or wireless (e.g., inductive coupling) connection with the charging unit **130** of the ring **405**.

(51) In some implementations, the environment **400** may include elements to facilitate harvesting of energy produced by motion of a user wearing the ring **405** (e.g., an embodiment of the ring **101**). For example, one or more permanent magnets may be disposed within the environment **400** (e.g., in the golf club **434**, the steering wheel **438**, or any other suitable location), configuring a magnetic field that the ring **405** may use for charging the battery **120**.

(52) The ring **405** may include (e.g., as a part of the charging unit **130**) a charging circuit that, in turn, includes an induction coil. When user motion changes magnetic flux from the one or more permanent magnets through the induction coil, a current may be induced in the coil. The charging circuit may be configured to charge a power source (e.g., the battery **120**) disposed within the ring **405** using the current induced in the coil.

IV. Example Diagram of an Electromagnetic Induction Charging System for a Smart Ring

(53) FIG. **5** is a schematic diagram of a charging system **500** for a smart ring **501** (e.g., smart ring **101**, smart ring **405**) that may operate, for example, within the environment **400**. The structure of the charging system **500** provides a general framework for implementations discussed in reference to FIG. **6**A, FIG. **6**B, FIG. **7**A, FIG. **7**B.

(54) Disposed within a housing **510** of the smart ring, may be a power source **520** (e.g., battery **120**), a charging circuit **530** (e.g., included in the charging unit **130**) that includes an induction coil **532** and rectification components **536**, and a smart-ring component **540** (referred to as component **540**). The component **540** may be a part or an entirety of the

controller **140**, the sensor unit **150**, the user input unit **170**, the communications unit **160** or the output unit **190**. The power source **520** may be in electrical connection with the charging circuit **530** and with the component **540**. The charging circuit **530** may be configured to charge (add energy to) the power source **520**, while the component **540** may be configured draw energy from the power source **520**. The component **540** may be further configured to perform at least one of: i) sense a physical phenomenon external to the housing **510**, ii) send communication signals to a communication device external to the housing **510**, or iii) implement a user interface, as discussed above. The system **500** also includes one or more permanent magnets disposed in a magnet module **550**. The magnet module **550** may be any aspect of the environment (e.g., in the golf club **434**, the steering wheel **438** of the environment **400**, in a keyboard of a computer, etc.) where one or more permanent magnets may be disposed to set up a magnetic field at the location of the ring. In certain examples, a magnet module **550** may be a part of the ring, as described in more detail below, in reference to FIG. **7**. Changing a relative position of the coil **532** and the magnet module **550** may change magnetic flux from the one or more permanent magnets of the module **550** through the induction coil **532**, generating electricity for charging the power source **520**. Changing the relative position of the coil **532** and the magnet module **550** may be the result of motion of a user wearing the ring **501**, for example.

(55) The induction coil **532** may have 1, 3, 10, 30, 100, 300, 1000, 3000, 10000 or any other suitable number of turns of a conductive wire (or another suitable conductor, such as a trace). The conductive wire may have a diameter of 10 μm, 20 μm, 50 μm, 100 μm, 200 μm, 500 μm, 1 mm, 2 mm, or any other suitable diameter. The conductive wire may be at least in part made of copper, gold, or any other suitable conductive material and may include polymer coating to electrically isolate neighboring turns. The permanent magnets of the magnet module **550** may be rare earth magnets (e.g., neodymium, samarium-cobalt), alnico magnets, or ceramic magnets of suitable dimensions.

(56) The rectifying components **536** included in the charging circuit **530** may be configured to convert the generally alternating current (AC) waveform generated in the coil **532** to a direct current (DC) output suitable for charging the power source **520**. Smoothing the voltage excursions in the generated AC (e.g., by converting to DC) may increase the energy efficiency of charging. The rectifying components **536** may include discrete or integrated electronic components configured as filters, diode-bridge rectifiers, H-bridge rectifiers, buck converters, boost converters, buck-boost converters, or any other suitable circuits. In some implementations, the rectifying components **536** may include a commutator.

(57) In a sense, the rectifying components **536** may be thought to include an rectifier configured to rectify or convert (e.g., in cooperation with a capacitor or another suitable component) AC to DC, thereby facilitating AC/DC conversion for current flowing between the coil **532** of the charging circuit and the remaining portion of the charging circuit. Depending on design considerations and various constraints, the rectifier may include a commutator or a solid-state rectifier. The commutator carries the advantage of rectifying the AC voltage across coil terminals without introducing a significant voltage drop and associated energy loss. The solid-state rectifier has fewer mechanical components than a commutator and may simplify the ring design, but may introduce a more significant voltage drop and energy loss than the commutator when rectifying.

V. Examples of the Induction Charging System for a Smart Ring

(58) FIG. **6**A illustrates an example charging system **600** (an embodiment of the charging system **500**) for a smart ring **605** (e.g., that may be ring **101**, ring **405**, or ring **501**) that includes an induction coil **606**. Advantageously, the charging system **600** may harvest the energy from the motion of the ring worn by a user while typing. The charging system **600** includes permanent magnets **620***a*, **620***b*, **620***c*, disposed in a wrist rest **625** which is disposed near or attached to a keyboard **630**.

(59) The induction coil **606** may have 1, 2, 5, 10, 20, 50, 100, 200, 500, 1000 or any other suitable number of turns of a conductive wire. The conductive wire may have a diameter of 10 μm, 20 μm, 50 μm, 100 μm, 200 μm, 500 μm, 1 mm, 2 mm, or any other suitable diameter. The conductive wire may be at least in part made of copper, gold, or any other suitable conductive material and may include polymer coating to electrically isolate neighboring turns.

(60) The permanent magnets **620***a*, **620***b*, **620***c* may be rare earth magnets (e.g., neodymium, samarium-cobalt), alnico magnets, ceramic magnets. Though FIG. **6**A illustrates three magnets **620***a*, **620***b*, **620***c*, the charging system **600** may use 1, 2, 5, 10, 20 or any other suitable number of magnets. The magnets **620***a*, **620***b*, **620***c* may have any suitable shapes, including, for example, cylinder, bar, block, or horseshoe. The magnets **620***a*, **620***b*, **620***c* may have any suitable dimensions. For example, the magnets **620***a*, **620***b*, **620***c* may be cylinders and have diameters of 2, 5, 10, 20 mm and heights of 2, 5, 10, 20 mm. The magnets **620***a*, **620***b*, **620***c* may be oriented in any suitable direction so as to have the magnetic flux density (magnetic field) gradients substantially maximized in a vertical (z-direction), a lateral (x-direction), or any other suitable direction. Generally, the magnets **620***a*, **620***b*, **620***c* may be disposed so as to substantially maximize magnetic flux variations through the coil **606** during expected movement patterns of the ring **605** worn by a user. At a working distance (e.g., 5 mm to 10 cm) from the magnets **620***a*, **620***b*, **620***c*, the magnetic field may vary from 20, 50, 100, 200, 500, 1000, 2000, 5000 gauss (or another suitable strength) close to the magnets to less than one gauss away from the magnets **620***a*, **620***b*, **620***c*.

(61) The induction coil **606** may have the turns wound substantially along the circumference of the ring **605**, as illustrated in FIG. **6**A, with the axis of the coil **606** aligning axially with respect to the ring **605**. An axis of the coil here may be defined as the direction substantially orthogonal to a conductive path (e.g., wire or trace) of the coil. In another implementation, illustrated in FIG. **6**B, one or more coils **607***a*, **607***b* may be disposed along the ring **605** with the axes oriented substantially radially with respect to the ring **605**. Furthermore, the coils **607***a*, **607***b* may be implemented as wires or as conductive material (e.g., gold, copper, etc.) traces (e.g., rectangular spiral traces) of any suitable width (20,

50, 100, 200, 500 μm, etc.) on a substrate. In some implementations, the substrate may be flexible.

(62) In general, coils **606** and **607***a*, **607***b* may electrically connect to a number of devices or components (e.g., rectifying components **536**) to form a charging circuit (e.g., the charging circuit **530**). The charging circuit may include one or more rectifiers (e.g., implemented with diodes) and filters (e.g., implemented with capacitors), as well as voltage regulators or DC/DC voltage converters. In implementations with multiple coils (e.g., coils **607***a*, **607***b*), each coil may be connected to a corresponding rectifier. The multiple coils may be connected in series or in parallel, depending on an implementation of the charging circuit.

(63) Returning to FIG. **6**A, user movement (e.g., typing on the keyboard) may change magnetic flux through the coil **606** by varying magnetic field strength in the vicinity of the ring **605**, or varying the orientation of the coil **606** with respect to the direction of the magnetic field. The charging circuit may then charge the power source (e.g., the power source **520**) using the electricity generated in the coil. In implementations with multiple coils (e.g., coils **607***a*, **607***b*), charging circuit may be configured to combine the electricity generated in each coil for charging the power source.

(64) FIG. **7**A illustrates an example implementation of the charging system **500** fully integrated into a self-charging smart ring **700**. The self-charging smart ring **700** may allow a user to charge the smart ring **700** by spinning a band, for example. The user may thereby fidget with the smart ring **700** recreationally while simultaneously charging the ring **700**. FIG. **7**B illustrates a section **701** of the smart ring **700** and, in more detail (and with a different perspective), components disposed within and a mechanical configuration of the smart ring **700**. The ring **700** includes a first band **702** and a second band **704** configured to rotate substantially coaxially with respect to the first band **702**. The second band **704** may include magnets **750***a*, **750***b*, **750***c*, **750***d*. In a sense, the second band **704** may be configured as the magnet module **550**. The first band **702**, on the other hand, may be configured as the housing **510**. Thus, in a sense, the smart ring **700** may be thought of as an implementation of the charging system **500** of FIG. **5**, where the magnet module **550** is movably attached to the ring housing **510**. The power source **520**, a coil **735** (e.g., coil **532**), rectifying components **536**, the component **540** may be disposed in the first band **702**.

(65) In some implementations, the magnets **750***a*, **750***b*, **750***c*, **750***d* may be distributed around the second band **704** at regular radial intervals with alternating dipole orientations in a substantially radial direction. That is, half of the magnets **750***a*, **750***b*, **750***c*, **750***d* (or any even number of magnets) may be oriented with the north pole facing the radially inner surface of the second band **704** and the other half of the magnets **750***a*, **750***b*, **750***c*, **750***d* may be oriented with the north pole facing the radially outer surface of the second band **704**. Alternating magnet orientation around the circumference of the second band **704** may enhance the changes in magnetic flux through the coil **735** when the bands **702**, **704** rotate with respect to each other.

(66) In some implementations, the first band **702** and the second band **704** may be radially disposed with respect to each other. An inner radius of at least a portion of the first band **702** may be larger than an outer radius of at least a portion of the second band **704**, with the first band **702** and the second band **704** forming, respectively, an outer portion of the ring **700** and an inner portion of the ring **700**. In certain examples, the first band **702** and the second band **704**, may form, respectively, the inner and the outer portions of the ring **700**. Still in other implementations, the first band **702** and the second band **704** may be substantially axially disposed with respect to each.

(67) In some implementations, the ring **700** may be configured as a radial ball bearing. To that end, the first band **702** and the second band **704** are configured as races for the radial ball bearing, in mechanical contact with each other by way of a suitable number of balls (e.g., ball **710**). The balls may be made of plastic, glass, ceramic, metal or metal alloy (e.g., steel) or any other suitable material. The bands **702**, **704** may include grooves **712**, **714** for confining the balls. The ring **700** may include a cage, a shield or other suitable radial ball bearing components. In some implementations, the ring **700** may be configured as a radial roller bearing, with the bands **702**, **702** configured as races for rollers.

(68) In operation, a user may apply a force to rotate the second band **704** with respect to the first band **702** (or, conversely, the first band **702** with respect to the second **704**), actuating a spin of the second band **704**. The ring **700** may be configured to reduce frictional losses (e.g., using bearings as discussed above), so as to remain spinning for a certain duration (e.g., several seconds or tens of seconds) between manual actuations by the user. As the second band **704** spins with respect to the first band **702**, magnetic flux through the coil **735** changes due to the motion of the magnets **750***a*, **750***b*, **750***c*, **750***d* with respect to the coil **735**, inducing an AC voltage across coil terminals. The rectifying components **536** may then convert the electricity into DC form suitable for charging the power source **520**.

(69) In some implementations, the rectifying components **536** may include a commutator. The commutator, for example, may mechanically reverse polarity of a connection between the coil **735** and the remaining portion of the charging circuit. The commutator may rectify the AC voltage across coil terminals without introducing a significant voltage drop and associated energy loss, as discussed above.

(70) In other implementations, solid state rectification components may be used instead of the commutator to reduce the number of mechanical component and simplify the ring design, but may introduce a voltage drop and associated energy loss.

(71) The second band **704** may be removably attached to the first band **702**. That is, removably attaching the second band **704** to the first band **702** may be done for the purpose of charging the smart ring **700**. Removably attaching here means that the user may attach or detach the second band **704** from the first band **702**. The detached second band may be removed from the finger. Removably attaching the second band **704** may preserve the rotational freedom of the second band **704** with respect to the first band. To that end, the second band **704** may include bearing balls, for example. The first

band **702** may be worn on the finger of the user without the second band **704** to enable some or all of the functionality (e.g., other than charging) of the smart ring **700**. In some implementations the removable second band **704** may be configured as a radially outer ring, as described above. In other implementations, the removable second band may be disposed axially (e.g., along the finger of the user wearing the smart ring **700**) proximal to the first band **702** in a side-by-side configuration. In the side-by-side configuration, magnets **750***a*, **750***b*, **750***c*, **750***d* disposed in the second band may be configured to have magnetic dipoles aligned in the substantially axial direction. Correspondingly, the coil **735** in the first band **702** may be aligned to maximize magnetic flux and magnetic flux changes induced by the magnets **750***a*, **750***b*, **750***c*, **750***d*. In any case, the removable second band **704** may allow reducing the bulk of the smart ring during the segments of operation when charging is not needed.

VI. Examples of Piezoelectric Charging of a Smart Ring

(72) Another technique for harvesting motion energy and converting it into electrical energy for use by a power source of a smart ring may include the use of a piezoelectric effect. The piezoelectric effect refers to the property of certain materials to generate voltage in response to mechanical stress that may produce microscopic deformations within the material. One advantage of using the piezoelectric effect is a relative ease of generating a substantial voltage (e.g., 1, 2, 5, 10, 20, 50, 100V or another suitable voltage), which may facilitate rectification.

(73) Example implementations of a smart ring configured for piezoelectric harvesting of mechanical energy for charging are discussed below with reference to FIG. **8**, FIG. **9**A, and FIG. **9**B. FIG. **8** is a schematic diagram of a smart ring **801**, analogous to the smart ring **501** described in reference to FIG. **5**. The smart ring **801** may represent any one or more of the smart rings **205**, **305**, **405**, **605**, **700**, or **900**. The configuration of the ring **801** may have the housing **510**, the power source **520**, and the component **540** that is configured to draw energy from the power source **520**, as described above. A charging circuit **830** may replace the charging circuit **530** in FIG. **5**. The charging circuit **830** may be configured to charge the power source **520** by way of electrical energy generated by a piezoelectric harvesting element **832**, as discussed above. Referring to the FIG. **5**, the piezoelectric harvesting element **832** may take the place of the induction coil **532**. The charging circuit **830** may like the charging circuit **530**, include rectification components **836**.

(74) The rectification components **836**, analogously to the rectification components **536** in FIG. **5**, may be configured to facilitate transforming the electrical energy generated by the piezoelectric harvesting element **832** into a form of electrical energy suitable for charging the power source **520**. Though the rectification components **836** may include similar elements to the rectification components **536**, e.g., diodes (e.g., in a bridge configuration), capacitors, etc., the rectification components **836** may be configured differently. More specifically, the rectification components **836** may be configured for the electrical characteristics of the piezoelectric harvesting element **832**, that may have high output voltage and high output impedance. For example, the rectification components **836** may include a buck converter for stepping down the voltage output of the piezoelectric harvesting element and improving power transfer efficiency.

(75) The piezoelectric harvesting element **832** may be disposed within the housing **510** of the smart ring **801** and configured for converting a mechanical energy input into electrical energy. The piezoelectric harvesting element **832** may be the only harvesting element in such a smart ring, or may be one of multiple harvesting elements. For example, the smart ring **801** may include the charging circuit **530** and the charging circuit **830** (which includes the piezoelectric harvesting element **832**) in an embodiment.

(76) The piezoelectric harvesting element **832** may include a deformable structure (e.g., a cantilever, a membrane, or another suitable structure) made at least in part from a piezoelectric material, the structure configured to deform under an application of an external force. The macroscopic deformation of the structure in response to the applied force may result in microscopic deformations within the piezoelectric material of the structure, and, subsequently, in voltage generation. A cumulative effect of multiple mechanical deformations, weather cyclical or aperiodic, of the piezoelectric material or structure may generate a voltage waveform that may be converted (e.g., by rectification) to drive charge accumulation in the power source **520**.

(77) After each mechanical deformation, the structure may substantially return to its original geometry, when the external force is no longer present, under the influence of the restorative force of the structure under deformation. In other implementations, the piezoelectric harvesting element may include an additional element (e.g., a spring) for providing a bias restorative force acting in opposition to the external force.

(78) A variety of piezoelectric materials may be adapted for use within the piezoelectric harvesting element **832**. The materials may be crystalline or amorphous and may include ceramics (e.g., lead zirconate titanate known as PZT, sodium potassium niobate, etc.), semiconductors (e.g., gallium nitride, zinc oxide, etc), and polymers (e.g., polyvyniladine chloride, polyvinyladine fluoride, etc.). The materials may be grown, deposited, or otherwise fabricated into thin or thick films or any other suitable form. The deformable structure (e.g., membrane, cantilever, etc.) of the piezoelectric harvesting element may include one or more piezoelectric materials.

(79) The deformable structure of the piezoelectric harvesting may be implemented as a microelectromechanical system (MEMS), using associated fabrication techniques. In other implementations, the deformable structure may be assembled from discrete mechanical components. Still in other implementations, the deformable structure may be implemented using additive manufacturing techniques, such as 3D printing. In certain examples, the deformable structure may be fabricated using a combination of suitable techniques.

(80) In some implementations, the deformable structure of the piezoelectric harvesting element **832** may be configured to vibrate or oscillate in response to a force impulse. A natural, e.g., resonant, frequency of oscillation of the structure may

depend, for example on the geometry and the distribution of mass of the deformable structure. The oscillation of the deformable structure may induce a voltage oscillation at the corresponding frequency. The oscillation of the deformable structure may facilitate harvesting energy from periodic external forces, e.g., vibrations, particularly those with frequencies close to the natural or resonant frequency of the deformable structure. It may be advantageous to configure one or more deformable structures of the piezoelectric harvesting element **832** to have resonant frequencies close to expected frequencies of external forces or vibrations. The piezoelectric harvesting element **832** may be configured to have a resonant frequency in the range of vibration frequencies of a steering wheel or motorcycle handlebars, for example, to facilitate charging the smart ring **801** via piezoelectric energy harvesting while driving. Generally, configuring the deformable structure to have a resonant response to a vibration source may facilitate charging when the ring **801** is mechanically coupled to the vibration source. A configuration may include the frequency of the source within a 3 dB bandwidth of the resonant response of the deformable structure.

(81) The piezoelectric harvesting element of the smart ring **801** may be configured to harvest energy from incidental movements of a user wearing the smart ring **801**. For example, the piezoelectric harvesting element may be configured to harvest energy from arm swings or finger tapping. To that end, a movable mass may be disposed within the housing **510** of the smart ring **801**. The movable mass may be mechanically coupled to the deformable structure of the piezoelectric harvesting element and exert a force on the deformable structure during acceleration associated with user movement and due to the inertia of the movable mass.

(82) It may be noted that the deformable structure of the piezoelectric harvesting element may be configured with a degree of freedom in its movement aligned with any suitable direction with respect to the smart ring **801**. For example, the movement direction of the deformable element may be substantially radial with respect to the ring-shaped housing **510**. In certain examples, the degree of movement may be in a tangential direction, moving along the circumference of the smart ring **801**. In certain examples, the degree of movement may be aligned with the axial direction, or substantially along the finger of the user. Furthermore, the directions of movement may be combined, a deformable element may be configured for two- or three-axis deformations, or multiple structures may be configured with different degrees of freedom, to take advantage of a variety of movements.

(83) In some implementations, the piezoelectric harvesting element may be configured to harvest energy from user movements that apply a force directly to the smart ring housing **510**, at least in part to charge the smart ring **801**. For example, the smart ring housing **510** may include a movable component that can translate a force, applied by the user, to the piezoelectric harvesting element. One suitable configuration may be analogous to the configuration of the smart ring discussed with reference to FIG. **7**A and FIG. **7**B. That is, in an example configuration, the smart ring **801** may include two bands configured to move with respect to each other to thereby generate mechanical energy for harvesting via the piezoelectric effect.

(84) FIG. **9**A and FIG. **9**B illustrate a smart ring **900** that includes two bands **902**, **904** configured to rotate with respect to one another for the purpose of generating mechanical energy for piezoelectric harvesting. The smart ring **900** is analogous to the smart ring **700** in FIG. **7**, in a sense that both allow a user to charge the smart rings **700**, **900** by rotating one of the movable bands **702**, **704**, **902**, **904**. The ring **900** may include a first band **902**, in which the piezoelectric harvesting element **832** may be disposed, and a second band **904**. The first band **902** and the second band **904** may be configured, respectively, as an inner ring and an outer ring, where the outer ring is disposed substantially outside the inner ring in a radial direction relative to the inner ring. The smart ring **900** may be configured as a radial ball bearing, as discussed above for the smart ring **700**.

(85) The second band **904** may include one or more actuation members **950***a-d*, disposed so as to cause a mechanical deformation within the piezoelectric harvesting element **832** that, in turn, generates electrical energy for harvesting. As the first and second bands **902**, **904** move with respect to one another, the repeated mechanical deformation contributes to charging the power source **520**.

(86) FIG. **9**B illustrates how the actuation member **950***a* causes the deformation in the piezoelectric harvesting element **832**. The piezoelectric harvesting element **832** may include a cantilever **960** that may have a protrusion or a catch that comes into contact with a corresponding protrusion on the actuation member **950***a*, as the actuation member **950***a* moves by the harvesting element **832**, the cantilever **960** deforms under the force applied by the actuation member **950***a*.

(87) The piezoelectric harvesting element may include one or more cantilevers (e.g., cantilever **960**) disposed along the circumference of the first band **902**. In some implementations, the one or more cantilevers may be oriented substantially in a tangential orientation with respect to the band **902**. In other implementations, the one or more cantilevers may be oriented radially with respect to the band **902**.

(88) Though the illustrated mechanism relies on mechanical contact between the actuation member **950***a* and the cantilever **960**, the piezoelectric harvesting element **832** may rely on other modes of actuation (e.g., magnetic, electrostatic). For example, the actuation member **950***a* may be a magnet, and the cantilever **960** may include paramagnetic or ferromagnetic material that is repelled or attracted by a passing magnet. Thus, as the user rotates the second band **904**, the magnets may cause intermittent deformations in the cantilevers (e.g., cantilever **960**) or other deformable structures of the piezoelectric harvesting element **832**, causing the generation of electrical energy for charging the power source **520**.

(89) It should be noted, that, though electromagnetic induction and piezoelectric effect are described in detail herein, other energy harvesting technologies may be integrated into a smart ring. Additional or alternative harvesting modules may

include triboelectric, thermoelectric, or photovoltaic charging, for example.

VII. Examples of Other Considerations

(90) When implemented in software, any of the applications, services, and engines described herein may be stored in any tangible, non-transitory computer readable memory such as on a magnetic disk, a laser disk, solid state memory device, molecular memory storage device, or other storage medium, in a RAM or ROM of a computer or processor, etc. Although the example systems disclosed herein are disclosed as including, among other components, software or firmware executed on hardware, it should be noted that such systems are merely illustrative and should not be considered as limiting. For example, it is contemplated that any or all of these hardware, software, and firmware components could be embodied exclusively in hardware, exclusively in software, or in any combination of hardware and software. Accordingly, while the example systems described herein are described as being implemented in software executed on a processor of one or more computer devices, persons of ordinary skill in the art will readily appreciate that the examples provided are not the only way to implement such systems.

(91) The described functions may be implemented, in whole or in part, by the devices, circuits, or routines of the system **100** shown in FIG. **1**. Each of the described methods may be embodied by a set of circuits that are permanently or semi-permanently configured (e.g., an ASIC or FPGA) to perform logical functions of the respective method or that are at least temporarily configured (e.g., one or more processors and a set instructions or routines, representing the logical functions, saved to a memory) to perform the logical functions of the respective method.

(92) Throughout this specification, some of the following terms and phrases are used.

(93) Communication Interface according to some embodiments: Some of the described devices or systems include a “communication interface” (sometimes referred to as a “network interface”). A communication interface enables the system to send information to other systems and to receive information from other systems, and may include circuitry for wired or wireless communication.

(94) Each described communication interface or communications unit (e.g., communications unit **160**) may enable the device of which it is a part to connect to components or to other computing systems or servers via any suitable network, such as a personal area network (PAN), a local area network (LAN), or a wide area network (WAN). In particular, the communication unit **160** may include circuitry for wirelessly connecting the smart ring **101** to the user device **104** or the network **105** in accordance with protocols and standards for NFC (operating in the 13.56 MHz band), RFID (operating in frequency bands of 125-134 kHz, 13.56 MHz, or 856 MHz to 960 MHz), Bluetooth (operating in a band of 2.4 to 2.485 GHz), Wi-Fi Direct (operating in a band of 2.4 GHz or 5 GHz), or any other suitable communications protocol or standard that enables wireless communication.

(95) Communication Link according to some embodiments: A “communication link” or “link” is a pathway or medium connecting two or more nodes. A link between two end-nodes may include one or more sublinks coupled together via one or more intermediary nodes. A link may be a physical link or a logical link. A physical link is the interface or medium(s) over which information is transferred, and may be wired or wireless in nature. Examples of physicals links may include a cable with a conductor for transmission of electrical energy, a fiber optic connection for transmission of light, or a wireless electromagnetic signal that carries information via changes made to one or more properties of an electromagnetic wave(s).

(96) A logical link between two or more nodes represents an abstraction of the underlying physical links or intermediary nodes connecting the two or more nodes. For example, two or more nodes may be logically coupled via a logical link. The logical link may be established via any combination of physical links and intermediary nodes (e.g., routers, switches, or other networking equipment).

(97) A link is sometimes referred to as a “communication channel.” In a wireless communication system, the term “communication channel” (or just “channel”) generally refers to a particular frequency or frequency band. A carrier signal (or carrier wave) may be transmitted at the particular frequency or within the particular frequency band of the channel. In some instances, multiple signals may be transmitted over a single band/channel. For example, signals may sometimes be simultaneously transmitted over a single band/channel via different sub-bands or sub-channels. As another example, signals may sometimes be transmitted via the same band by allocating time slots over which respective transmitters and receivers use the band in question.

(98) Memory and Computer-Readable Media according to some embodiments: Generally speaking, as used herein the phrase “memory” or “memory device” refers to a system or device (e.g., the memory unit **144**) including computer-readable media (“CRM”). “CRM” refers to a medium or media accessible by the relevant computing system for placing, keeping, or retrieving information (e.g., data, computer-readable instructions, program modules, applications, routines, etc.). Note, “CRM” refers to media that is non-transitory in nature, and does not refer to disembodied transitory signals, such as radio waves.

(99) The CRM may be implemented in any technology, device, or group of devices included in the relevant computing system or in communication with the relevant computing system. The CRM may include volatile or nonvolatile media, and removable or non-removable media. The CRM may include, but is not limited to, RAM, ROM, EEPROM, flash memory, or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to store information, and which can be accessed by the computing system. The CRM may be communicatively coupled to a system bus, enabling communication between the CRM and other systems or components coupled to the system bus. In some implementations the CRM may be coupled to the system bus via a memory interface (e.g., a memory controller). A

memory interface is circuitry that manages the flow of data between the CRM and the system bus.
(100) Network according to some embodiments: As used herein and unless otherwise specified, when used in the context of system(s) or device(s) that communicate information or data, the term “network” (e.g., the networks **105** and **440**) refers to a collection of nodes (e.g., devices or systems capable of sending, receiving or forwarding information) and links which are connected to enable telecommunication between the nodes.
(101) Each of the described networks may include dedicated routers responsible for directing traffic between nodes, and, in certain examples, dedicated devices responsible for configuring and managing the network. Some or all of the nodes may be also adapted to function as routers in order to direct traffic sent between other network devices. Network devices may be inter-connected in a wired or wireless manner, and network devices may have different routing and transfer capabilities. For example, dedicated routers may be capable of high volume transmissions while some nodes may be capable of sending and receiving relatively little traffic over the same period of time. Additionally, the connections between nodes on a network may have different throughput capabilities and different attenuation characteristics. A fiberoptic cable, for example, may be capable of providing a bandwidth several orders of magnitude higher than a wireless link because of the difference in the inherent physical limitations of the medium. If desired, each described network may include networks or sub-networks, such as a local area network (LAN) or a wide area network (WAN).
(102) Node according to some embodiments: Generally speaking, the term “node” refers to a connection point, redistribution point, or a communication endpoint. A node may be any device or system (e.g., a computer system) capable of sending, receiving or forwarding information. For example, end-devices or end-systems that originate or ultimately receive a message are nodes. Intermediary devices that receive and forward the message (e.g., between two end-devices) are also generally considered to be “nodes.”
(103) Processor according to some embodiments: The various operations of example methods described herein may be performed, at least partially, by one or more processors (e.g., the one or more processors in the processor unit **142**). Generally speaking, the terms “processor” and “microprocessor” are used interchangeably, each referring to a computer processor configured to fetch and execute instructions stored to memory. By executing these instructions, the processor(s) can carry out various operations or functions defined by the instructions. The processor(s) may be temporarily configured (e.g., by instructions or software) or permanently configured to perform the relevant operations or functions (e.g., a processor for an Application Specific Integrated Circuit, or ASIC), depending on the particular embodiment. A processor may be part of a chipset, which may also include, for example, a memory controller or an I/O controller. A chipset is a collection of electronic components in an integrated circuit that is typically configured to provide I/O and memory management functions as well as a plurality of general purpose or special purpose registers, timers, etc. Generally speaking, one or more of the described processors may be communicatively coupled to other components (such as memory devices and I/O devices) via a system bus.
(104) The performance of certain of the operations may be distributed among the one or more processors, not only residing within a single machine, but deployed across a number of machines. In some example embodiments, the processor or processors may be located in a single location (e.g., within a home environment, an office environment or as a server farm), while in other embodiments the processors may be distributed across a number of locations.
(105) Although specific embodiments of the present disclosure have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the present disclosure is not to be limited by the specific illustrated embodiments.

Claims

1. A smart ring comprising: a housing configured to be worn by a user; a power source disposed within the housing; a charging circuit disposed within the housing, wherein the charging circuit includes a harvesting element configured to charge the power source, and wherein the harvesting element is configured to harvest energy from a motion of a portion of the user wearing the housing; and a device at least partially disposed within the housing, wherein the device is configured to draw energy from the power source.
2. The smart ring of claim 1, wherein: the harvesting element comprises a piezoelectric harvesting element.
3. The smart ring of claim 1, wherein: the harvesting element comprises an induction coil.
4. The smart ring of claim 3, wherein: the induction coil includes one or more turns along a circumference of a ring-shaped housing.
5. The smart ring of claim 1, wherein: the harvesting element includes one or more additively manufactured components.
6. The smart ring of claim 1, wherein: the motion of the portion of the user comprises an arm swing.
7. The smart ring of claim 1, wherein: the harvesting element is configured to vibrate or oscillate in response to the motion of the portion of the user wearing the housing.
8. A smart ring comprising: a means for housing, wherein the means for housing is configured to be worn by a user; a means for powering, wherein the means for powering is disposed within the means for housing; a means for charging disposed within the means for housing, wherein the means for charging includes a means for harvesting configured to charge the means for powering, and wherein the means for harvesting is configured to harvest energy from a motion of a portion of the user wearing the housing; and a means for drawing energy from the means for powering, wherein the means for drawing energy from the means for powering is at least partially disposed within the means for housing.

9. The smart ring of claim 8, wherein: the means for harvesting is a means for harvesting piezoelectric energy.
10. The smart ring of claim 8, wherein: the means for harvesting comprises an induction coil.
11. The smart ring of claim 10, wherein: the induction coil includes one or more turns along a circumference of a ring-shaped housing.
12. The smart ring of claim 8, wherein: the means for harvesting includes one or more additively manufactured components.
13. The smart ring of claim 8, wherein: the motion of the portion of the user comprises an arm swing.
14. A method for operating a smart ring, the method comprising: generating electrical energy at a charging circuit within the smart ring, wherein the charging circuit includes a harvesting element, and wherein the harvesting element is configured to harvest energy from a motion of a portion of a user wearing the smart ring; charging a power source disposed within the smart ring by delivering the electrical energy from the charging circuit to the power source; and causing the power source to provide power to a device disposed within the smart ring.
15. The method of claim 14, wherein: the harvesting element comprises a piezoelectric harvesting element.
16. The method of claim 14, wherein: the harvesting element comprises an induction coil.
17. The method of claim 16, wherein: the induction coil includes one or more turns along a circumference of the smart ring.
18. The method of claim 14, wherein: the harvesting element includes one or more additively manufactured components.
19. The method of claim 14, wherein: the motion of the portion of the user comprises an arm swing.
20. The method of claim 14, wherein: the harvesting element is configured to vibrate or oscillate in response to the motion of the portion of the user wearing the smart ring.